

Chapter 4

Permutations and Combinations

Solutions

SECTION - A

Objective Type Questions (One option is correct)

[Permutations]

1. If ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800$, then r is
(1) 39 (2) 41 (3) 28 (4) 43

Sol. Answer (2)

$$\Rightarrow \frac{{}^{56}P_{r+6}}{{}^{54}P_{r+3}} = 30800$$

$$\Rightarrow \frac{56! (54 - r - 3)!}{(56 - r - 6)! \times 54!} = 30800$$

$$\Rightarrow \frac{56 \times 55 (51 - r)!}{(50 - r)!} = 30800$$

$$\Rightarrow \frac{(51 - r)(50 - r)!}{(50 - r)!} = 10$$

$$\Rightarrow 51 - r = 10$$

$$\Rightarrow r = 41$$

2. If ${}^nC_x = 56$ and ${}^nP_x = 336$, then n is equal to

- (1) 7 (2) 8 (3) 6 (4) 10

Sol. Answer (2)

$${}^nP_x = x! {}^nC_x$$

$$\Rightarrow x! \times 56 = 336$$

$$\Rightarrow x! = \frac{336}{56} = 6$$

$$\Rightarrow x! = 1 \times 2 \times 3$$

$$\Rightarrow x! = 3!$$

$$\therefore x = 3$$

Now, ${}^nC_3 = 56$

$$\Rightarrow \frac{n!}{3!(n-3)!} = 56$$

$$\Rightarrow \frac{n(n-1)(n-2)(n-3)!}{(n-3)!} = 3! \times 56$$

$$\Rightarrow n(n-1)(n-2) = 6 \times 7 \times 8$$

$$\therefore n = 8$$

3. In how many ways the letters of the word RAINBOW be arranged?

- (1) 5040 (2) 4050 (3) 8040 (4) 6050

Sol. Answer (1)

There are 7 letters in the word RAINBOW and each letter is used only once. So, all the 7 letters can be arranged in 7! ways.

4. How many numbers greater than 50000 can be formed with the digits 4, 5, 6, 7 and 8 if no digit being repeated?

- (1) 96 (2) 256 (3) 218 (4) 126

Sol. Answer (1)

Ten thousands place cannot assume 4 since required numbers are greater than 50000.

$$\therefore \text{Required arrangements} = 4 \times {}^4P_4 \\ = 4 \times 24 = 96$$

5. In how many ways can 10 soldiers stand in two rows having 5 soldiers in each row?

- (1) 3628800 (2) 7257600 (3) 35400 (4) 985000

Sol. Answer (1)

Any 5 soldiers can stand in one row and the remaining 5 soldiers can stand in other row.

$$\therefore \text{Required numbers of arrangements} = {}^{10}P_5 \times {}^5P_5 \\ = 3628800$$

6. If the maximum number of trials required to open all locks when there are n locks and n keys is 105, then $n =$

- (1) 13 (2) 14 (3) 35 (4) 20

Sol. Answer (2)

To open the first lock, maximum number of trial is n .

Similarly to open subsequent locks, maximum number of trials is $n-1, n-2, \dots$

$$\text{Maximum number of trials} = n + (n-1) + (n-2) + \dots + 1 = \frac{n(n+1)}{2} = 105$$

$$n(n+1) = 14 \times 15$$

$$\therefore n = 14.$$

7. How many arrangements can be made out of the letters of the word COMMITTEE, taken all at a time, such that the four vowels do not come together?

- (1) 12600 (2) 45600 (3) 43200 (4) 216

Sol. Answer (3)

Number of letter in the word COMMITTEE is 9 out of these M occurs twice, T occurs twice, E occurs twice and remaining are different.

$$\therefore \text{Number of arrangement} = \frac{9!}{2! 2! 2!} = 45360$$

Now, there are 4 vowels O, I, E, E in the given word, taking them as one letter, we have to arrange 6 letters which include 2M's and 2T's.

$$\therefore \text{Number of arrangements} = \frac{6!}{2! 2!} = 180 \text{ ways.}$$

$$\text{Number of arrangements of 4 vowels} = \frac{4!}{2!} = 12$$

$$\therefore \text{Number of arrangements in which four vowels always come together} = 180 \times 12 = 2160.$$

$$\begin{aligned} \text{Hence, required number of arrangements} &= 45360 - 2160 \\ &= 43200 \end{aligned}$$

8. How many different words can be made with the letters of the word NAINITAL such that each of the word begins with L and end with T?

- (1) 85 (2) 88 (3) 92 (4) 90

Sol. Answer (4)

When L and T are fixed as first and last letters of the word, then we have, only 6 letters to be arranged.

$$\text{Hence, required number of arrangements} = \frac{6!}{2! 2! 2!} = 90$$

9. How many numbers greater than a million can be formed with the digits 5, 5, 2, 2, 1, 7, 6?

- (1) 1320 (2) 1180 (3) 1000 (4) 1260

Sol. Answer (4)

For a number greater than million, we have to use all 7 seven digits which include 5 twice, 2 twice and remaining are different.

$$\therefore \text{Number of arrangements} = \frac{7!}{2! 2!} = 1260$$

10. How many different signals can be given using any number of flags from 5 flags of different colours?

- (1) 325 (2) 2030 (3) 1888 (4) 1920

Sol. Answer (1)

Number of signals using single flag is 5P_1

Number of signals using two different flags is 5P_2

Number of signals using three different flags is 5P_3

Number of signals using four different flags is 5P_4

Number of signals using five different flags is 5P_5

$$\begin{aligned} \therefore \text{Required number different signals} &= {}^5P_5 + {}^5P_4 + {}^5P_3 + {}^5P_2 + {}^5P_1 \\ &= 120 + 120 + 60 + 20 + 5 \\ &= 325 \end{aligned}$$

11. The number of permutations of n different objects taken k at a time, when repetitions are allowed is

(1) k^n (2) n^k (3) $n!$ (4) nP_k

Sol. Answer (2)

Number of permutations of n different objects, taken k at a time, when repetitions are allowed is given by n^k .

12. Number of natural numbers not exceeding 4321 can be formed with the digits 1, 2, 3, 4 if repetition is allowed is

(1) 123 (2) 113 (3) 313 (4) 222

Sol. Answer (3)

Total one digit number = 4

Total two digit numbers = $4 \times 4 = 16$

Total three digit numbers = $4 \times 4 \times 4 = 64$

Number of 4 digit numbers starting with 1 = $4 \times 4 \times 4 = 64$

Number of 4 digit numbers starting with 2 = $4 \times 4 \times 4 = 64$

Number of 4 digit numbers starting with 3 = $4 \times 4 \times 4 = 64$

Number of 4 digit numbers starting with 41 = $4 \times 4 = 16$

Number of 4 digit numbers starting with 42 = $4 \times 4 = 16$

Number of 4 digit numbers starting with 431 = 4

Number of 4 digit numbers starting with 432 = 1

Total number of 4 digit numbers = $3 \times 64 + 2 \times 16 + 4 + 1$
= 229

Hence, required number of natural number not exceeding 4321 is $4 + 16 + 64 + 229 = 313$

13. Number of ways in which the letters of the word RAINBOW be arranged such that N and B are together is

(1) 2560 (2) 1540 (3) 540 (4) 1440

Sol. Answer (4)

Number of letters in the word RAINBOW = 7

Suppose N and B as single letter, then number of remaining letters is 6, which can be arranged themselves in $6!$ ways. Also N and B are arranged in $2!$ ways.

$\therefore$ Required number of arrangements = $6! \times 2!$
= 1440

14. Number of even numbers greater than 300 that can be formed with the digits 1, 2, 3, 4, 5 such that no digit being repeated is

(1) 111 (2) 900 (3) 600 (4) 800

Sol. Answer (1)

A number will be even number if its unit digit is 2 or 4. Number greater than 300 may be of 3-digit numbers, 4-digit numbers or 5-digit numbers.

Case-I : 3-digit numbers

When 2 is fixed at unit place, we have 3, 4 and 5 i.e., three digits for filling hundreds place and remaining 3-digits for filling tens place. Similarly, when 4 is fixed at the unit place, hundred place can be filled by 2 choices and the ten's digit can be filled by 3 choices.

$\therefore$ Number of 3-digit numbers = $9 + 6$
= 15

Case-II : 4-digit numbers

For unit digit, available choice is 2 and the remaining digits can be arranged in 4P_3 ways.

$$\begin{aligned}\text{Number of 4-digit numbers} &= 2 \times {}^4P_3 \\ &= 48\end{aligned}$$

Case-III : 5-digit numbers

Unit digit, can be arranged in 2 ways and the remaining digits can be arranged in 4P_4 ways.

$$\begin{aligned}\text{Number of 5-digit numbers} &= 2 \times {}^4P_4 \\ &= 48\end{aligned}$$

Hence, total number of numbers greater than 300 is $15 + 48 + 48 = 111$.

15. There are 6 books of physics, 3 of chemistry and 4 of biology. Number of ways in which these books be placed on a shelf if the books of the same subject are to be together is

- (1) 622080 (2) 888000 (3) 222000 (4) 413080

Sol. Answer (1)

Since, the same subject are to placed together

$\therefore$ Number of arrangements of placing Physics = $6! = 720$

Number of arrangements of placing Chemistry = $3! = 6$

Number of arrangements of placing Biology = $4! = 24$

Set of 6 books of Physics, 3 of Chemistry and 4 of Biology can arranged themselves in $3!$ ways.

$$\begin{aligned}\text{Required number of arrangements of sets of books} &= 720 \times 6 \times 24 \times 3! \\ &= 622080\end{aligned}$$

16. A person has 5 shirts, 4 coats and 7 ties. Number of ways in which he can wear them is

- (1) 100 (2) 140 (3) 96 (4) 124

Sol. Answer (2)

$$\begin{aligned}\text{Number of arrangements} &= 5 \times 4 \times 7 \\ &= 140\end{aligned}$$

17. Number of ways in which 15 different books can be arranged on a shelf so that two particular books shall not be together is

- (1) $14 \times 15!$ (2) $13 \times 14!$ (3) $14! \times 15!$ (4) $(15!)^2$

Sol. Answer (2)

Number of arrangements of 15 different books = $15!$

Let two particular books as a set of one book, then remaining number of different books is 14, which can arranged in $14!$. Now, the set of 2 books also arranged themselves in $2!$ ways.

$$\begin{aligned}\therefore \text{Required number of arrangements in which a set of two particular books shall not be together} &= 15! - 14! \times 2! \\ &= 15 \times 14! - 14! \times 2! \\ &= 14! (15 - 2) \\ &= 13 \times 14!\end{aligned}$$

18. How many 6 digit numbers can be formed out of the digits of the number 113226?

(1) 156

(2) 180

(3) 280

(4) 120

Sol. Answer (2)

$$\text{Number of arrangements} = \frac{6!}{2!2!} = 180$$

19. Number of three digit numbers such that at least one of the digits is 9, if repetitions allowed, is

(1) 252

(2) 648

(3) 864

(4) 468

Sol. Answer (1)

$$\begin{aligned}\text{Number of 3 digit numbers} &= 9 \times 10 \times 10 \\ &= 900\end{aligned}$$

$$\begin{aligned}\text{Number of 3 digit numbers excluding 9} &= 8 \times 9 \times 9 \\ &= 648\end{aligned}$$

$$\begin{aligned}\therefore \text{Number of 3 digit numbers such that at least one of the digits is 9} \\ &= 900 - 648 \\ &= 252\end{aligned}$$

20. Number of ways in which the letters of the word TAMANNA be arranged is

(1) 120

(2) 420

(3) 840

(4) 500

Sol. Answer (2)

Number of letters in the word TAMANNA = 7. Out of these there are 3A's and 2N's and the remaining letters are different.

$$\therefore \text{Required number of arrangements} = \frac{7!}{3!2!} = 420$$

21. Number of three letter words that can be formed using only vowels but each only once is

(1) 100

(2) 50

(3) 60

(4) 80

Sol. Answer (3)

Number of vowels = 5

$$\therefore \text{Required number of words} = {}^5P_3 = 60$$

22. Number of 3 digit numbers that can be formed having unit digit as zero and repetition of digit is allowed, is

(1) 72

(2) 81

(3) 100

(4) 90

Sol. Answer (4)

$$\begin{aligned}\text{Number of 3 digit numbers} &= 9 \times 10 \times 1 \\ &= 90\end{aligned}$$

23. Number of different words that can be made using the letters of the word HALLUCINATION if all consonants are together is

(1) $\left(\frac{7!}{2!2!}\right)^2$

(2) $\frac{7!}{2!2!}$

(3) $\frac{14!}{7!}$

(4) $\frac{14}{2!2!}$

Sol. Answer (1)

Number of letters in the word HALLUCINATION = 13

Number of consonants = 7

Out of these there are 2 L's, 2 N's and remaining are different.

Number of arrangements of consonants = $\frac{7!}{2!2!}$

Let set of consonants as single letter, then remaining letters in the word HALLUCINATION is 7, in which there are 2 A's, 2 I's and the remaining are different

$\therefore$ Number of arrangements = $\frac{7!}{2!2!}$

Hence, required number of arrangements of the word HALLUCINATION if all consonants are together

$$= \frac{7!}{2!2!} \times \frac{7!}{2!2!} = \left(\frac{7!}{2!2!} \right)^2$$

24. Number of ways in which the letters of the word SUCCESSFUL be arranged is

(1) $\frac{10!}{5!}$

(2) $\frac{13!}{5!}$

(3) $\frac{10!}{3!(2!)^2}$

(4) $\frac{10!}{(3!2!)^2}$

Sol. Answer (3)

Required number of arrangements = $\frac{10!}{3!2!2!}$

25. Number of 6 digit numbers that can be formed using the digit 2 two times and the digit 5 four times is

(1) 16

(2) 15

(3) 24

(4) 18

Sol. Answer (2)

Required number of arrangements = $\frac{6!}{2!4!} = 15$

26. Number of ways in which 15 billiard balls be arranged in a row if 3 are identical red, 7 are identical white and remaining are identical black balls is

(1) $15!$

(2) $\frac{15!}{7!3!5!}$

(3) $7!3!5!$

(4) $\frac{15!}{7!5!}$

Sol. Answer (2)

Required number of arrangements = $\frac{15!}{3!5!7!}$

27. Number of 7 digit telephone numbers that can be formed from the digits 0, 1, 2,9, if each telephone number begins with digit 5 is

(1) 6^{10}

(2) 10^6

(3) $10!$

(4) $^{10}C_6$

Sol. Answer (2)

Each digit can be filled by any one of the 10 digits.

$\therefore$ Required number of telephone numbers = 10^6

28. Four dice are rolled. The number of possible out comes in which atleast one die show 6 is

- (1) 671 (2) 168 (3) 176 (4) 650

Sol. Answer (1)

Number of faces on a single die = 6

Number of dice = 4

$$\therefore \text{Number of possible outcomes when four dice are rolled} = 6 \times 6 \times 6 \times 6 \\ = 6^4$$

$$\text{Let none of dice show 6, then number of possible outcomes} = 5 \times 5 \times 5 \times 5 \\ = 5^4$$

$$\text{Required number of outcomes} = 6^4 - 5^4 \\ = 671$$

29. The total number of permutations of n different objects taken at least one but not more than r at a time, where each object may be repeated any number of time is

- (1) $\frac{n(n^n - 1)}{n - 1}$ (2) $\frac{n^r - 1}{n - 1}$ (3) $\frac{n(n^r - 1)}{n - 1}$ (4) $\frac{n(n^r - 1)}{r - 1}$

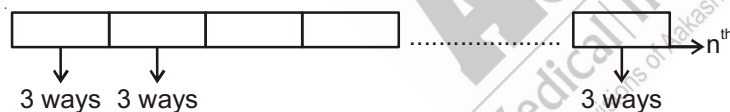
Sol. Answer (3)

$$\text{Required number} = n + n^2 + \dots + n^r = \frac{n.(n^r - 1)}{n - 1}$$

30. An n -digit number is a positive number with exactly n digits. Nine hundred distinct n digit numbers are to be formed using only three digit 5, 7, 9. The smallest value of n for which this is possible is

- (1) 6 (2) 7 (3) 8 (4) 9

Sol. Answer (2)



$$3^n \geq 900$$

$$\Rightarrow n = 7$$

31. The number of arrangements of the letter of the word PAPAYA in which the two 'P' do not appear adjacently is

- (1) 40 (2) 60 (3) 80 (4) 36

Sol. Answer (1)

PAPAYA

$$\text{Total number of arrangements without restriction} = \frac{6!}{3!2!} = 60$$

$$\text{Total number of arrangements when both (P) are together} = \frac{5!}{3!} = 20$$

Total number of arrangement when no two 'P' are together
= $60 - 20 = 40$

32. How many different nine digit numbers can be formed with the number 223355888 by rearranging its digits so that the odd digits occupy even positions?

(1) 16 (2) 36 (3) 60 (4) 180

Sol. Answer (3)

223355888

No. of even positions = 4

No. of odd positions = 5

$$\text{Total no. of arrangement} = \frac{4!}{2!2!} \times \frac{5!}{2!3!} = 6 \times 10 = 60.$$

33. How many words are formed if the letters of the word GARDEN are arranged with the vowels in alphabetical order?

(1) 120 (2) 240 (3) 360 (4) 480

Sol. Answer (3)

Vowels are A, E

In half of arrangements, A will come before E and in half of arrangement E will come before A.

$$\therefore \text{Number of ways in which vowels will in alphabetical order} = \frac{6!}{2} = \frac{720}{2} = 360$$

34. If the letters of the word SACHIN are arranged in all possible ways and these words are written out as in dictionary, then the word SACHIN appears at serial number

(1) 602 (2) 603 (3) 600 (4) 601

Sol. Answer (4)

Alphabetical order of word SACHIN is ACHINS

Total no. of word beginning from A, C, H, I, N is $5 \times 5 \times 4 \times 3 \times 2 \times 1$
= 600

SACHIN is the 1st word beginning from 'S'.

$\therefore$ Rank of SACHIN is 601.

[Combinations]

35. In how many ways can a committee of 4 men and 3 women be appointed from 6 men and 8 women?

(1) 480 (2) 308 (3) 840 (4) 640

Sol. Answer (3)

4 men can be selected out of 6 men in 6C_4 ways and 3 women can be selected out of 8 women in 8C_3 ways.

$$\therefore \text{Required number of selections of committee} = {}^6C_4 \times {}^8C_3 = 840$$

36. In how many ways 7 members forming a committee out of 11 be selected so that 3 particular members must be included?

(1) 81 (2) 70 (3) 95 (4) 125

Sol. Answer (2)

If 3 particular members are included, then remaining 4 members are selected out of the 8 members. It can be done in 8C_4 ways.

$${}^8C_4 = \frac{8!}{4!4!} = 70$$

37. ${}^{15}C_9 + {}^{15}C_8 - {}^{15}C_6 - {}^{15}C_7$ equals to

- (1) 0 (2) 1 (3) ${}^{15}C_8$ (4) None of these

Sol. Answer (1)

$$\begin{aligned} & {}^{15}C_9 + {}^{15}C_8 - {}^{15}C_6 - {}^{15}C_7 \\ &= {}^{15}C_9 + {}^{15}C_8 - ({}^{15}C_7 + {}^{15}C_6) \\ &= {}^{16}C_9 - {}^{16}C_7 \quad [\because {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r] \\ &= {}^{16}C_7 - {}^{16}C_7 \quad [{}^nC_r = {}^nC_{n-r}] \\ &= 0 \end{aligned}$$

38. The number of words which can be formed out of the letters of the word PARTICLE, so that vowels occupy the even place is

- (1) 7! (2) ${}^4C_4 \times {}^3C_3$ (3) 180 (4) $4! \times 5!$

Sol. Answer (4)

Number of letters in the word PARTICLE = 8

Number of vowels = 3

3 vowels can be arranged at 4 even places in 4P_3 ways. After arranging the vowels, remaining consonants can be arranged themselves in 5! ways.

$$\begin{aligned} \therefore \text{Required number of arrangements} &= {}^4P_3 \times 5! \\ &= 4! \times 5! \end{aligned}$$

39. The number of ways in which three girls and ten boys can be seated in two vans, each having numbered seats, three in the front and four at the back is

- (1) ${}^{14}C_{12}$ (2) ${}^{14}P_{13}$ (3) 14! (4) 10^3

Sol. Answer (2)

Total number of seats = $2(4 + 3) = 14$

Total number of persons = $3 + 10 = 13$

$$\therefore \text{Number of required arrangements} = {}^{14}P_{13}$$

40. Number of ways in which the letters of the word MOBILE be arranged so that the consonants always occupy the odd places is

- (1) 18 (2) 54 (3) 36 (4) 72

Sol. Answer (3)

Number of letters in the word MOBILE = 6

Number of consonants = 3

Number of odd places = 3

Now, 3 consonants can be arranged in 3P_3 ways. After arranging the 3 consonants 3 vowels can be arranged in remaining places in 3P_3 ways.

$$\therefore \text{Required number of arrangements} = {}^3P_3 \times {}^3P_3 \\ = 36$$

41. Number of different words that can be formed from 15 consonants and 5 vowels by taking 2 consonants and 4 vowels in each word is

(1) ${}^{15}P_2 \times {}^5C_4$ (2) ${}^{15}C_2 \times {}^5C_4 \times 6!$ (3) ${}^{15}P_2 \times {}^5P_4$ (4) ${}^{15}P_2 \times {}^5P_4 \times 2! \times 4!$

Sol. Answer (2)

2 consonants can be selected out of 15 consonants in ${}^{15}C_2$ ways. Similarly, 4 vowels can be selected out of 5 vowels in 5C_4 ways.

$\therefore$ Selection of 6 letters is in ${}^{15}C_2 \times {}^5C_4$ ways

Now, these six selected letters arranged themselves in $6!$ ways.

$\therefore$ Required number of words = ${}^{15}C_2 \times {}^5C_4 \times 6!$

42. A box contains 7 red, 6 white and 4 blue balls. Number of ways of selection of three red balls is

(1) 35 (2) 45 (3) 27 (4) 36

Sol. Answer (1)

Only red balls is selected

Number of total red balls = 7

Number of selected red balls = 3

$$\therefore \text{Required number of selection} = {}^7C_3 \\ = 35$$

43. In a test paper there are 10 questions. Number of ways in which 6 questions to be answered is

(1) 105 (2) 210 (3) 310 (4) 220

Sol. Answer (2)

Required number of ways is ${}^{10}C_6$

44. In a meeting everyone had shaken hands with everyone else, it was found that 66 handshakes were exchanged.

Number of persons present in the meeting is

(1) 17 (2) 12 (3) 13 (4) 18

Sol. Answer (2)

One handshake is performed if two persons handshake to each other. If number of persons be n , then number of handshakes is nC_2 .

Now, ${}^nC_2 = 66$

$$\Rightarrow n(n-1) = 132$$

$$\Rightarrow n(n-1) = 12 \times 11$$

$$\therefore n = 12.$$

Hence, number of persons is 12

45. Number of all permutations of the set of four letters E, O, S, P taken two at a time is

- (1) ${}^4C_2 \times 2!$ (2) 4C_2 (3) $2!$ (4) $4!$

Sol. Answer (1)

$$\begin{aligned}\text{Required number of permutations} &= {}^4P_2 \\ &= {}^4C_2 \times 2! \quad [\because {}^nP_r = r! {}^nC_r]\end{aligned}$$

46. Number of ways in which 5 prizes be given away to 8 students when each student is eligible for all the prizes is

- (1) 5^8 (2) 8^5 (3) $8 \times 5!$ (4) $5 \times 8!$

Sol. Answer (2)

Each student get any of the 5 prizes

$\therefore$ Number of ways in which prizes is distributed = 8^5 .

47. In a badminton tournament each player played one game with all the other players. Number of players participated in the tournament if they played 105 games is

- (1) 35 (2) 15 (3) 12 (4) 10

Sol. Answer (2)

Let number of players = n

$\therefore$ Number of games played = nC_2

According to the question,

$${}^nC_2 = 105$$

$$\Rightarrow n(n-1) = 210$$

$$\Rightarrow n(n-1) = 14 \times 15$$

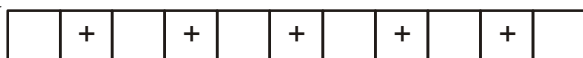
$$\Rightarrow n = 15.$$

48. Number of ways in which 5 plus (+) signs and 5 minus (−) signs be arranged in a row so that no two minus signs are together is

- (1) 6 (2) 7 (3) 8 (4) 10

Sol. Answer (1)

Fix any one the 5 plus (+) signs or 5 minus (−) signs as shown in the figure. Let it be + signs. Since all are identical, therefore they can be arranged in a single way.



After arranging the plus (+) signs, minus (−) signs can be arranged in 6 blank box in $\frac{6!}{5!}$ ways

[$\therefore$ Minus signs are also identical]

$$= 6$$

$\therefore$ Total number of arrangements = $6 \times 1 = 6$.

49. Find the number of ways in which a person can buy 6 chocolates, if there are three types of chocolates available. (Chocolates of same type are identical)

(1) 10 (2) 28 (3) 3^6 (4) 6^3

Sol. Answer (3)

Each chocolate can be selected in 3 ways. i.e., $3 \times 3 \times 3 \times 3 \times 3 \times 3 = 3^6$ ways

50. ${}^nC_{r+1} + {}^nC_{r-1} + 2 \cdot {}^nC_r =$

(1) ${}^{n+2}C_{r+1}$ (2) ${}^{n+1}C_r$ (3) ${}^{n+1}C_{r+1}$ (4) ${}^{n+2}C_r$

Sol. Answer (1)

$${}^nC_{r+1} + {}^nC_{r-1} + 2 \cdot {}^nC_r$$

$${}^nC_r + {}^nC_{r+1} + {}^nC_{r-1} + {}^nC_r = {}^{n+1}C_{r+1} + {}^{n+1}C_r = {}^{n+2}C_{r+1}$$

51. In a party of 30 people each shakes hands with the others. Number of hand shakes took place in the party is

(1) 435 (2) 870 (3) 900 (4) 535

Sol. Answer (1)

For each selection of two persons, number of hand shake is 1

$$\therefore \text{Total number of hand shakes} = {}^{30}C_2 = \frac{30 \times 29}{2} = 435.$$

52. On the eve of new year each student of class XI sends cards to his classmates. If there are 40 students in the class number of cards exchanged during the process is

(1) 1200 (2) 600 (3) 780 (4) 1560

Sol. Answer (4)

For each selection of two students number of cards exchanged is 2.

$$\therefore \text{Total number of cards} = {}^{40}C_2 \times 2 = 40 \times 39 = 1560$$

53. A box contains 3 red, 4 white and 2 black balls. The number of ways in which 3 balls can be drawn from the box, so that at least one ball is red, is (all balls are different)

(1) 45 (2) 64 (3) 84 (4) 85

Sol. Answer (2)

Number of ways of selecting 3 balls without any restriction = 9C_3 .

Number of ways when no red ball is selected = 6C_3 .

$$\text{Required number of ways} = {}^9C_3 - {}^6C_3 = 64.$$

54. In a certain test there are n questions. In this test 2^{n-k} students gave wrong answers to at least k questions, where $k = 1, 2, \dots, n$. If the total number of wrong answers given is 2047, then n equals

(1) 8 (2) 10 (3) 11 (4) 15

Sol. Answer (3)

The number of answering exactly r questions wrongly is $2^{n-r} - 2^{n-r-1}$

The number of students answering r questions wrongly is r

$\therefore$ Total number of wrong answers

$$= 1(2^{n-1} - 2^{n-2}) + 2(2^{n-2} - 2^{n-3}) + \dots + 2^0 \cdot n$$

$$\text{Total number of wrong answers} = 2^{n-1} + 2^{n-2} + 2^{n-3} + \dots + 2^0$$

$$= \frac{2^n - 1}{2 - 1} = 2047$$

$$\Rightarrow 2^n - 1 = 2047 \Rightarrow 2^n = 2048 = 2^{11}$$

$$\Rightarrow n = 11$$

55. In how many ways 6 students and 4 teachers be arranged in a row so that no two teachers are together?

- (1) 604800 (2) 246800 (3) 258600 (4) 55500

Sol. Answer (1)

First of all the 6 students can be arranged in $6!$ ways then 4 teachers can be arranged in 7 places in 7P_4 ways.

$$\begin{aligned} \therefore \text{Required number of arrangements} &= 6! \times {}^7P_4 \\ &= 720 \times 840 = 604800 \end{aligned}$$

56. Eight chairs are numbered 1 to 8. Two women and three men wish to occupy one chair each. First the women choose the chairs from amongst the chairs marked 1 to 4; and then the men select the chairs from amongst the remaining. The number of possible arrangements is

- (1) ${}^6C_3 \times {}^4C_2$ (2) ${}^4P_2 \times {}^6P_3$ (3) ${}^4C_2 + {}^4C_3$ (4) ${}^4P_2 + {}^4P_3$

Sol. Answer (2)

Women will occupy 2 chairs from first 4 chairs in ${}^4C_2 2!$ ways

Now men will occupy 3 chairs in ${}^6C_3 3!$

$$\text{Required number of ways} = {}^6C_2 2! \times {}^6C_3 \times 3! = {}^4P_2 \times {}^6P_3$$

57. Harsha invites 13 guests to a dinner and places 8 of them at one table and remaining 5 at the other, the table being round. The number of ways he can arrange the guests is

- (1) $\frac{11!}{40}$ (2) $9!$ (3) $\frac{13!}{40}$ (4) $\frac{12!}{40}$

Sol. Answer (3)

For first table, 8 guest can be selected and arranged in ${}^{13}C_8 \times 7!$ ways.

Now remaining 5 on another table can be arranged in $4!$ ways

$$\therefore \text{Required number of ways} = {}^{13}C_8 \times 7! \times 4! = \frac{13!}{8!5!} \times 7! \times 4! = \frac{13!}{40}$$

58. If one quarter of all the subsets containing three elements of the integers 1, 2, 3, ..., m contain the integer 5, then m is equal to

- (1) 12 (2) 10 (3) 14 (4) 11

Sol. Answer (1)

Number of 3 element subsets = mC_3

Number of 3 element subset which contains 5 = ${}^{m-1}C_2$

According to question ${}^mC_3 \times \frac{1}{4} = {}^{m-1}C_2$

$$\Rightarrow {}^mC_3 = 4 \cdot {}^{m-1}C_2$$

$$\frac{m!}{3!(m-3)!} = 4 \cdot \frac{m-1!}{2!(m-3)!} = \frac{m}{6} = \frac{4}{2}$$

$$\Rightarrow m = 12$$

[Geometric Problems]

59. The number of parallelograms that can be formed from a set of four parallel lines intersecting another set of five parallel lines is

- (1) 60 (2) 30 (3) 48 (4) 64

Sol. Answer (1)

Parallelogram can be formed if a pair of parallel lines and another pair of parallel lines can be selected.

$$\therefore \text{Required number of parallelograms} = {}^4C_2 \times {}^5C_2 = 60$$

60. Number of different straight lines that can be formed by joining 12 different points on a plane of which 4 are collinear is

- (1) 16 (2) 61 (3) 65 (4) 37

Sol. Answer (2)

Total number of lines formed by 12 points = ${}^{12}C_2$

Number of lines formed by 4 points = 4C_2

$\therefore$ Required number of lines = Total lines formed by 12 points – number of lines formed by 4 collinear points + 1

$$= {}^{12}C_2 - {}^4C_2 + 1 = 61$$

61. A polygon has 90 diagonals. Number of its sides is

- (1) 25 (2) 17 (3) 15 (4) 14

Sol. Answer (3)

Number of diagonals in n -sided polygon = Number of total lines – Number of sides

$$= {}^nC_2 - n$$

According to the question,

$$\Rightarrow {}^nC_2 - n = 90$$

$$\Rightarrow \frac{n(n-1)}{2} - n = 90$$

$$\Rightarrow n = 15$$

62. Number of triangles that can be formed by joining the 10 non-collinear points on a plane is

- (1) 136 (2) 140 (3) 128 (4) 120

Sol. Answer (4)

A triangle can be formed by joining 3 non-collinear points. Hence number of triangles is equal to the number of ways in which 3 points can be selected out of 10 non-collinear points.

$$\therefore \text{Number of triangles} = {}^{10}C_3 \\ = 120$$

63. There are 12 points in a plane in which 6 are collinear. Number of different triangles that can be drawn by joining them is

- (1) 220 (2) 200 (3) 240 (4) 320

Sol. Answer (2)

$$\text{Required number of triangles} = {}^{12}C_3 - {}^6C_3 = 200.$$

64. Let T_n denotes the number of triangles which can be formed by using the vertices of a regular polygon of n sides. If $T_{n+1} - T_n = 21$, then n is equal to

- (1) 5 (2) 7 (3) 6 (4) 4

Sol. Answer (2)

$$T_{n+1} - T_n = 21$$

$${}^{n+1}C_3 - {}^nC_3 = 21$$

$$\Rightarrow n = 7.$$

65. There are m points on the line AB and n points on the line AC , excluding the point A . Triangles are formed joining these points

- (i) When point A is not included,
 (ii) When point A is included.

The ratio of the number of such triangles is

- (1) $\frac{m+n+2}{2}$ (2) $\frac{m+n-2}{2}$ (3) $\frac{m+n-2}{m+n}$ (4) $\frac{m+n+2}{m+n}$

Sol. Answer (3)

When A is not included

$$\text{Number of triangles} = {}^mC_2 \cdot {}^nC_1 + {}^mC_1 \cdot {}^nC_2 = \frac{mn(m+n-2)}{2}$$

When A is included

$$\text{Number of triangles} = \frac{mn(m+n-2)}{2} + {}^mC_1 \cdot {}^nC_1 = \frac{mn(m+n)}{2}$$

$$\text{Required ratio} = \frac{m+n-2}{m+n}$$

66. The number of rectangles, of any size excluding squares from the rectangle of size 8×7 is

- (1) 784 (2) 840 (3) 896 (4) None of these

Sol. Answer (2)

$$\text{Required number} = {}^9C_2 \times {}^8C_2 - \sum_{r=1}^7 (9-r)(8-r)$$

$$= 36 \times 28 - 72 \times 7 - \left(\frac{7(7+1)(15)}{6} \right) + \frac{17(7)(7+1)}{2}$$

$$= 1008 - 504 - 140 + 476 = 840$$

[Miscellaneous]67. Number of ways in which n balls be randomly distributed in n cells is

- (1) $n!$ (2) $n(n+1)$ (3) n^n (4) 2^n

Sol. Answer (3)One of the balls out of n balls is selected and it can be put any one of the n cells. $\therefore$ First ball can be put in a cell n ways.Similarly, remaining each ball can also be put in n ways.

$$\therefore \text{Total number of arrangements} = \underbrace{n \times n \times n \times \dots \times n}_{n\text{-times}} = n^n$$

68. There are four balls of different colours and four boxes of colours, same as those of balls. The number of ways in which the balls, one each in one box, could be placed such that any ball does not go to the box of its own colour, is

- (1) 9 (2) 13 (3) 8 (4) 16

Sol. Answer (1)

$$\text{Required number of ways} = 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right) = 12 - 4 + 1 = 9.$$

69. Ramya gives a dinner party for 5 guests. The number of ways in which they may be selected from among 9 friends if two of the friends will not attend the party together is

(1) 84 (2) 133 (3) 91 (4) 126

Sol. Answer (3)

$$\text{Total number of ways} = {}^9C_5 - {}^7C_3 = 91$$

(7C_3 = Number of ways when two particular friends appear together).

70. At an election, a voter may vote for any number of candidates not greater than the number to be elected. There are 10 candidates and 4 are to be elected. If a voter votes for at least one candidate, then the number of ways in which he can vote is

(1) 6210 (2) 385 (3) 1110 (4) 5040

Sol. Answer (2)

A vote can voter maximum 4 votes and minimum 1 vote

$$\begin{aligned}\therefore \text{Required number of ways} &= {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_3 + {}^{10}C_4 \\ &= 10 + 45 + 120 + 210 = 385\end{aligned}$$

71. The number of ways in which a composite number $N = 2^2 3^2 5^2 7^2$ can be resolved into two factors which are prime to each other is

(1) 2 (2) 2^2 (3) 2^3 (4) 2^4

Sol. Answer (3)

$$N = 2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2$$

Prime divisors = 4 (2, 3, 5, 7)

$$\therefore \text{Required number} = 2^{4-1} = 2^3$$

72. The number of divisors of 9600, including 1 and 9600 are

(1) 50 (2) 48 (3) 58 (4) 46

Sol. Answer (2)

$$9600 = 2^7 \times 3^1 \times 5^2$$

Total number of divisors = Total number of ways in which 2, 3 or 5 can be related

$$= (7 + 1)(1 + 1)(2 + 1) = 8 \times 2 \times 3 = 48.$$

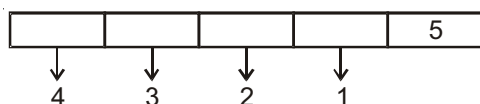
73. The sum of all five digit numbers that can be formed using the digits 1, 2, 3, 4, 5 when repetition of digits is not allowed is

(1) 366000 (2) 660000 (3) 360000 (4) 3999960

Sol. Answer (4)

5 will occur at unit place = $4 \times 3 \times 2 = 24$ times.

Similarly, 1, 2, 3, 4, 5 will occur at each place 24 times.



$$\begin{aligned}
 \therefore \text{Required sum} &= (1 + 2 + 3 + 4 + 5) \times 24 \times 10^4 \\
 &+ (1 + 2 + 3 + 4 + 5) \times 24 \times 10^3 + (1 + 2 + 3 + 4 + 5) \times 24 \times 10^2 + (1 + 2 + 3 + 4 + 5) \times 24 \times 10^1 + \\
 &(1 + 2 + 3 + 4 + 5) \times 24 \times 10^0 \\
 &= 24 \times 15 (11111) \\
 &= 3999960
 \end{aligned}$$

74. The number of ways of selecting 8 books from a library which has 9 identical books each on Mathematics, Physics, Chemistry and English is

(1) 165 (2) 166 (3) 167 (4) 168

Sol. Answer (1)

$$\text{Required number of ways} = \text{coeff. of } x^8 \text{ in } (1 + x + x^2 + \dots + x^9)^4$$

$$= {}^{8+4-1}C_{4-1} = {}^{11}C_3 = 165$$

75. Let y be an element of the set $A = \{1, 2, 3, 5, 6, 10, 15, 30\}$ and x_1, x_2, x_3 be integers such that $y = x_1 x_2 x_3$. Then the number of positive integral solutions of $x_1 x_2 x_3 = y$ is

(1) 64 (2) 27 (3) 81 (4) 30

Sol. Answer (1)

$$\text{When } y = 1 \quad x_1 \cdot x_2 \cdot x_3 = 1 \quad \text{Only one solutions} = x_1 = x_2 = x_3 = 1$$

$$y = 2 \quad 1 \times 1 \times 2 \quad \text{no. of solutions} = \frac{3!}{2!} = 3$$

$$y = 3 \quad 1 \times 1 \times 3 \quad \text{no. of solutions} = \frac{3!}{2!} = 3$$

$$y = 5 \quad 1 \times 1 \times 5 \quad \text{no. of solutions} = \frac{3!}{2!} = 3$$

$$y = 6 \quad 1 \times 2 \times 3 \quad \text{no. of solutions} = 3! + \frac{3!}{2!} = 9$$

$$\text{or } 1 \times 1 \times 6$$

$$y = 10 \quad 1 \times 2 \times 5 \quad \text{no. of solutions} = 3! + \frac{3!}{2!} = 9$$

$$\text{or } 1 \times 1 \times 10$$

$$y = 15 \quad 1 \times 3 \times 5 \quad \text{no. of solutions} = 3! + \frac{3!}{2!} = 9$$

$$\text{or } 1 \times 1 \times 15$$

$$y = 30 \quad 1 \times 5 \times 6 \quad \text{no. of ways} = 4 \times 3! + \frac{3!}{2!}$$

$$= 24 + 3 = 27$$

$$\text{or } 2 \times 3 \times 5$$

$$\text{or } 1 \times 3 \times 10$$

$$\text{or } 1 \times 15 \times 2$$

$$\text{or } 1 \times 1 \times 30$$

$\therefore$ Total number of ways = 64.

76. The number of divisors of the form $4k + 2$, $k \geq 0$ of the integer 240 is

- (1) 4 (2) 8 (3) 10 (4) 3

Sol. Answer (1)

$$240 = 2^4 \times 3 \times 5$$

Number of divisors of the form : $4k + 2$, $k \geq 0$ or $2(2k + 1)$

Such divisors are 2.1 or 2.3 or 2.5 or 2.15.

So number of such divisors = 4.

77. The exponent of 12 in $100!$ is

- (1) 10 (2) 12 (3) 13 (4) 48

Sol. Answer (4)

Exponent of 2 in $100!$

$$= \left[\frac{100}{2} \right] + \left[\frac{100}{2^2} \right] + \left[\frac{100}{2^3} \right] + \left[\frac{100}{2^4} \right] + \left[\frac{100}{2^5} \right] + \left[\frac{100}{2^6} \right] = 50 + 25 + 12 + 6 + 3 + 1 = 97$$

$$= \left[\frac{100}{3} \right] + \left[\frac{100}{3^2} \right] + \left[\frac{100}{3^3} \right] + \left[\frac{100}{3^4} \right] = 33 + 11 + 3 + 1 = 48$$

$$\text{Now, } 100! = 2^{97} 3^{48} 5^{97} 7^c = (12)^{48} \cdot 2.5^{97} 7^c$$

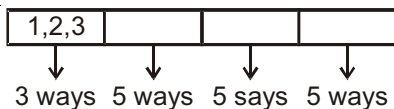
$\therefore$ Exponent of 12 is 48.

78. How many numbers greater than 1000 but not greater than 4000 can be formed from the digits 0, 1, 2, 3, 4, repetition of the digits being allowed?

- (1) 374 (2) 375 (3) 376 (4) 377

Sol. Answer (2)

Required number of ways = $3 \times 5^3 - 1 + 1 = 375$ (One is subtracted to exclude 1000 and one is added to include 4000).



79. The number of ways of distributing 12 different objects among three persons such that one particular person is always getting 6 objects and the remaining two persons are getting three objects each, is

- (1) $\frac{12!}{6!(3!)^2 2!}$ (2) $\frac{12!}{6!(3!)^2}$ (3) $\frac{12!}{(6!)^2}$ (4) $\frac{(12)!}{(3!)^2 (6!)^2}$

Sol. Answer (2)

$$\text{Required number of ways} = {}^{12}C_6 \times {}^6C_3 \times {}^3C_3 = \frac{12!}{6!(3!)^2}$$

80. The number of all five digit numbers which are divisible by 4 that can be formed from the digits 0, 1, 2, 3, 4 without repetition is

(1) 36

(2) 30

(3) 34

(4) 40

Sol. Answer (2)

			4	0	6
			2	0	6
			0	4	6
			1	2	4
			2	4	6
			2	2	4

Last two digits should be divisible by 4.

$\therefore 30$

81. The total number of ways in which four boys and four girls can be seated around a round table, so that no two girls sit together is equal to

(1) $7!$ (2) $3!4!$ (3) $4!$ (4) $4!4!$

Sol. Answer (2)

The number of ways in which all boys can be sit around a round table is $3!$ and hence they create and such voids, and they can be arranged in $4!$ ways = $3!4!$.

82. There are 15 couples taking part in a single tournament, the number of ways in which they can be paired such that no two real lives couple play in the same team is

(1) $15! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots - \frac{1}{15!} \right)$

(2) $15!$ (3) ${}^{30}C_2$ (4) ${}^{30}C_2 - {}^{15}C_1$

Sol. Answer (1)

$$15! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots - \frac{1}{15!} \right)$$

Clearly this is equal to the number of dearrangement.

83. The number of different ways of assigning 10 marks to 3 questions is

(1) 72

(2) 71

(3) 36

(4) 84

Sol. Answer (3)

Any question cannot be of less than 1 mark

Clearly the number of different ways

$$= {}^{10-1}C_{3-1}$$

$$= {}^9C_2 = \frac{9 \times 8}{2 \times 1} \Rightarrow 36$$

84. There are 3 apartments A , B and C for rent in a building. Each apartment will accept either 3 or 4 occupants. The number of ways of renting the apartments to 10 students
- (1) 12600 (2) 10800 (3) 13500 (4) 15000

Sol. Answer (1)

This is clearly a case of "a four and two threes". Now A can be given 4 occupants, and B and C , 3 each in ${}^{10}C_4 \cdot {}^6C_3 \cdot {}^3C_3$ ways.

The total number will 3 times this

$$\begin{aligned}\text{Hence, required number is} &= 3 \cdot {}^{10}C_4 \cdot {}^6C_3 \cdot {}^3C_3 \\ &= 12600\end{aligned}$$

85. A committee of 12 is to be formed from 9 women and 8 men. Then the number of ways of selecting committees so that men are in majority is
- (1) 3202 (2) 1134 (3) 2702 (4) 1314

Sol. Answer (2)

For women to be in majority following selection can be done.

No. of women	No of men	No of ways
9	3	${}^9C_9 \times {}^8C_3$
8	4	${}^9C_8 \times {}^8C_4$
7	5	${}^9C_7 \times {}^8C_5$
		2702

For men to be in majority.

No. of women	No. of men	No. of ways
4	8	${}^9C_4 \times {}^8C_8$
5	7	${}^9C_5 \times {}^8C_7$
		1134

86. A double-decker bus has 5 empty seats in the upper deck and 5 empty seats in the lower deck. 10 people board the bus of which 2 are old people and 3 are children. The children refuse to take seats lower deck while old people insist to stay lower deck. How many different sitting arrangements of the people are possible?
- (1) 121000 (2) 81000 (3) 10000 (4) 144000

Sol. Answer (4)

We can select any 2 seats down stair and give seat to old person this can be done in $= {}^5C_2 \times 2!$ ways

Again we can select any 3 seats upstairs and give it to children this can be done in $= {}^5C_3 \times 3!$ ways

Rest can be seated in $5!$ ways

$$\begin{aligned}\therefore \text{Required number of ways} &= {}^5C_2 \times 2! \times {}^5C_3 \times 3! \times 5! \\ &= 20 \times 60 \times 120 = 144000\end{aligned}$$

87. The letters of the word **COCHIN** are permuted and all the permutations are arranged in an alphabetical order as in a English dictionary. The number of words that appear before the word **COCHIN** is [IIT-JEE 2007]

(1) 360 (2) 192 (3) 96 (4) 48

Sol. Answer (3)

Arrange the letters of the word COCHIN alphabetical order. Clearly the number of words before the word is **96**.

88. The number of seven digit integers, with sum of the digits equal to 10 and formed by using the digits 1, 2, and 3 only, is [IIT-JEE 2009]

(1) 55 (2) 66 (3) 77 (4) 88

Sol. Answer (3)

Since sum of digits of seven digit numbers formed by the digits 1, 2 and 3 is 10, hence following cases may arise:

Case I :

$$1 + 1 + 1 + 1 + 1 + 2 + 3 = 10$$

$$\text{Number of possible seven digit numbers} = \frac{7!}{5!} = 42$$

Case II :

$$1 + 1 + 1 + 1 + 2 + 2 + 2 = 10$$

$$\text{Number of possible seven digit numbers} = \frac{7!}{3!4!} = 35$$

$$\text{Total number of seven digit numbers} = 42 + 35 = 77$$

89. Let $S = \{1, 2, 3, 4\}$. The total number of unordered pairs of disjoint subsets of S is equal to [IIT-JEE 2010]

(1) 25 (2) 34 (3) 42 (4) 41

Sol. Answer (4)

Let

The number of elements in $A : 0$ Choices for $B : 2^4$

The number of elements in $A : 1$ Choices for $B : 38$

The number of elements in $A : 2$ Choices for $B : 3$

Total number of possible subsets is 41

90. The total number of ways in which 5 balls of different colours can be distributed among 3 persons so that each person gets at least one ball is [IIT-JEE 2012]

(1) 75 (2) 150 (3) 210 (4) 243

Sol. Answer (2)

x	y	z	
3	1	1	$\longrightarrow 3 \times {}^5C_3 \times {}^2C_1 \times {}^1C_1 = 60$
2	2	1	$\longrightarrow 3 \times {}^5C_2 \times {}^3C_2 \times {}^1C_1 = 90$
			<u>Total = 150</u>

91. Six cards and six envelopes are numbered 1, 2, 3, 4, 5, 6 and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same number and moreover the card numbered 1 is always placed in envelope numbered 2. Then the number of ways it can be done is [JEE(Advanced)-2014]

(1) 264 (2) 265 (3) 53 (4) 67

Sol. Answer (3)

$$\frac{6! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} \right)}{5}$$

$$= \frac{265}{5} = 53$$

92. Let $S = \{1, 2, 3, \dots, 9\}$. For $k = 1, 2, \dots, 5$, let N_k be the number of subsets of S , each containing five elements out of which exactly k are odd. Then $N_1 + N_2 + N_3 + N_4 + N_5 =$ [JEE(Advanced)-2017]

(1) 125 (2) 210 (3) 252 (4) 126

Sol. Answer (4)

Required number of subsets

$$\begin{aligned} &= {}^5C_1 \times {}^4C_4 + {}^5C_2 \times {}^4C_3 + {}^5C_3 \times {}^4C_2 + {}^5C_4 \times {}^4C_1 + {}^5C_5 \times {}^4C_0 \\ &= 5 + 40 + 60 + 20 + 1 \\ &= 126 \end{aligned}$$

Alternate method

Coefficient of x^5 in $(1+x)^5(1+x)^4$

$$\begin{aligned} &= {}^9C_5 \\ &= 126 \end{aligned}$$

SECTION - B

Objective Type Questions (More than one options are correct)

1. If nC_4 , nC_5 and nC_6 are in A.P., then the value of n is
(1) 14 (2) 11 (3) 7 (4) 8

Sol. Answer (1, 3)

$$2 \times {}^nC_5 = {}^nC_4 + {}^nC_6$$

$${}^nC_5 - {}^nC_4 = {}^nC_6 - {}^nC_5$$

On solving we get the answer.

2. The results of 11 chess matches (as win, lose or draw) are to be forecast. Out of all possible forecasts, the number of ways in which 8 correct and 3 incorrect results can be forecast is
(1) ${}^{11}C_8 \times 8$ (2) $2^3 \times {}^{11}C_3$ (3) 1320 (4) $8! \times 3!$

Sol. Answer (1, 2, 3)

$$\text{Required number of ways} = {}^{11}C_8 \times 2^3 = {}^{11}C_3 \times 2^3 = 1320$$

3. Ritu has to answer 10 out of 13 questions in an examination. The number of ways in which she can answer if she must answer at least 3 of the first five questions is

(1) 276 (2) ${}^{13}C_{10} - {}^5C_3$
 (3) ${}^5C_3 \cdot {}^8C_7 + {}^5C_4 \cdot {}^8C_6 + {}^5C_5 \cdot {}^8C_5$ (4) ${}^{13}C_{10} - {}^5C_2$

Sol. Answer (1, 2, 3, 4)

$$\begin{aligned}\text{Required number of ways} &= {}^5C_3 \cdot {}^8C_7 + {}^5C_4 \cdot {}^8C_6 + {}^5C_5 \cdot {}^8C_5 \\ &= {}^{13}C_{10} - {}^5C_2 = {}^{13}C_{10} - {}^5C_3 = 276\end{aligned}$$

4. Let N denote the number of ways in which n boys can be arranged in a line so that 3 particular boys are separated then, then N is equal to

(1) ${}^{n-2}P_3(n-3)!$ (2) $\frac{(n-2)!(n-3)!}{n-5!}$ (3) ${}^{n-2}C_3 3!(n-3)!$ (4) $(n-3)(n-4)(n-2)!$

Sol. Answer (1, 2, 3, 4)

First of all we arrange $(n-3)$ boys by $\frac{n-3}{1}$ ways. Then between $(n-2)$ gaps between $(n-3)$ boys 3 particular boys can be arranged by ${}^{n-2}P_3$ ways

$$\text{Total number of ways} = {}^{n-2}P_3 \times (n-3)!$$

5. There are 25 students in a class. The teacher takes 5 of them at a time, to zoological garden as often as he can, without taking the same 5 students more than once. Then the number of visits, the teacher makes to the garden exceeds that of a student by

(1) ${}^{25}C_5 - {}^{24}C_5$ (2) ${}^{24}C_4$ (3) ${}^{24}C_5$ (4) ${}^{25}C_5 - {}^{24}C_4$

Sol. Answer (3, 4)

$$\text{Number of visits the teacher makes to the garden} = {}^{25}C_5$$

$$\text{Number of visits the student makes to the garden} = {}^{24}C_4$$

$$\text{Required number of ways} = {}^{25}C_5 - {}^{24}C_4 = {}^{24}C_5$$

6. The number of ways in which 20 girls be seated round a table if there are only 10 chairs

(1) $\frac{{}^{20}C_{10}}{2}$ (2) ${}^{20}C_{10} \cdot 9!$ (3) $\frac{{}^{20}C_{10}}{10}$ (4) $\frac{{}^{20}P_{10}}{10}$

Sol. Answer (2, 4)

$$\text{Required no. of ways} = {}^{20}C_{10} \times 9! = \frac{{}^{20}P_{10}}{10}$$

7. The number of non-negative integral solutions of $x_1 + x_2 + x_3 + x_4 \leq n$ (where n is a non-negative integer) is

(1) ${}^{n+3}C_3$ (2) ${}^{n+4}C_4$ (3) ${}^{n+3}C_n$ (4) ${}^{n+4}C_n$

Sol. Answer (2, 4)

$$x_1 + x_2 + x_3 + x_4 \leq n$$

$\therefore$ Number of non-negative integral solutions

$$= {}^{n+4-1}C_{4-1} + {}^{n-1+4-1}C_{4-1} + {}^{4-2+4-1}C_{4-1} + \dots + {}^{0+4-1}C_{4-1}$$

$$= {}^{n+3}C_3 + {}^{n+2}C_3 + {}^{n+1}C_3 + \dots + {}^3C_3 = {}^{n+4}C_4$$

Alternate method

$$x_1 + x_2 + x_3 + x_4 + x_5 = n, \quad 0 \leq x_5 \leq n$$

Number of solutions

$$= {}^{n+5-1}C_{5-1} = {}^{n+4}C_4$$

8. If N is the number of positive integral solutions of $x_1 x_2 x_3 x_4 = 770$, then

- (1) N is divisible by 4 distinct primes
(2) N is a perfect square
(3) N is a perfect fourth power
(4) N is a perfect 8th power

Sol. Answer (2, 3, 4)

$$770 = 2 \times 5 \times 11 \times 7 = x_1 x_2 x_3 x_4$$

2 can be distributed in 4 ways, similarly 5, 11 and 7 can be distributed in 4 ways respectively.

$$\text{Required number of ways} = 4 \cdot 4 \cdot 4 = 4^3 = N$$

9. There are n white and n red balls marked 1, 2, 3, n . The number of ways we can arrange these balls in a row so that neighbouring balls are of different colours is

- (1) $2(n)!$
(2) $(2n)!$
(3) $2(n!)^2$
(4) $2 \cdot \frac{(2n)!}{2^n C_n}$

Sol. Answer (3, 4)

$$\text{Required number of ways} = 2 \cdot n! \cdot n! = 2(n!)^2 = 2 \cdot \frac{(2n)!}{2^n C_n}$$

(Either red will come first or white).

10. There are 15 bulbs in a room. Each one of them can be operated independently. The number of ways in which the room can be lighted is

- (1) $8^5 - 1$
(2) $(32)^2 - 1$
(3) $(32)^3 - 1$
(4) $8^4 - 1$

Sol. Answer (1, 3)

Each bulb will be either 'on' or 'off'

$$\text{Required number of ways} = 2^{15} - 1 = (32)^3 - 1 = (8)^5 - 1.$$

11. The number of ways in which 5 distinct toys can be distributed among 8 children is

- (1) 2^{15}
(2) 8^5
(3) 5^8
(4) 32768

Sol. Answer (1, 2, 4)

$$\text{Required number of ways is} = 8 \times 8 \times 8 \times 8 \times 8 = 8^5$$

12. The number of six digit numbers that can be form from the digits 1, 2, 3, 4, 5, 6 and 7. So that the digits do not repeat and the terminal digits are even is

- (1) $5.4! \cdot 3!$
(2) 72
(3) 720
(4) $5! \cdot 3!$

Sol. Answer (1, 3, 4)

$$\text{Required number} = ({}^3C_2 \times 2!) \times ({}^5C_4 \times 4!) = 720$$

13. The number of integers which lie between 1 and 10^6 and which have sum of the digits equal to 12 is

- (1) ${}^{17}C_5 - 6 \times {}^7C_5$
(2) 5382
(3) 6062
(4) 8055

Sol. Answer (1, 3)

The number of ways in which the sum of digits will be equal to 12 is

$$= \text{Coefficient of } x^{12} \text{ in } (x^0 + x^1 + x^2 + \dots + x^9)^6$$

$$= \text{Coefficient of } x^{12} \text{ in } \left(\frac{1-x^{10}}{1-x} \right)^6$$

$$= \text{coefficient of } x^{12} \text{ in } (1-x^{10})^6 (1-x)^{-6}$$

$$= \text{coefficient of } x^{12} \text{ in } (1-x)^{-6} (1 - {}^6C_1 \cdot x^{10} + \dots)$$

$$= \text{coefficient of } x^{12} \text{ in } (1-x)^{-6} - {}^6C_1 \cdot \text{coefficient of } x^2 \text{ in } (1-x)^{-6}$$

$$= {}^{12+6-1}C_{6-1} - {}^6C_1 \cdot {}^{2+6-1}C_{6-1}$$

$$= {}^{17}C_5 - {}^{6 \cdot 7}C_5 = 6062$$

Hence answer is (1, 3)

14. $\sum_{r=0}^m {}^{n+r}C_n$ is equal to

(1) ${}^{n+m+1}C_{n+1}$

(2) ${}^{n+m+2}C_n$

(3) ${}^{n+m+1}C_m$

(4) ${}^{n+m-1}C_{m-1}$

Sol. Answer (1, 3)

$$\sum_{r=0}^m {}^{n+r}C_n = \sum_{r=0}^m {}^{n+r}C_r$$

$$= {}^nC_0 + {}^{n+1}C_1 + {}^{n+2}C_2 + \dots + {}^{n+m}C_m$$

$$= 1 + n + 1 + {}^{n+2}C_2 + \dots + {}^{n+m}C_m$$

$$= (n + 2 + {}^{n+2}C_2) + {}^{n+2}C_3 + \dots + {}^{n+m}C_m$$

$$= ({}^{n+2}C_1 + {}^{n+2}C_2) + {}^{n+3}C_3 + \dots + {}^{n+m}C_m$$

$$= ({}^{n+3}C_2 + {}^{n+3}C_3) + {}^{n+4}C_4 + \dots + {}^{n+m}C_m$$

$$= {}^{n+m+1}C_m$$

15. Between two junction stations A and B there are 12 intermediate stations. The number of ways in which a train can be made to stop at 4 of these stations so that no two of these halting station are consecutive is

(1) 9C_4

(2) ${}^{12}C_4 - 4$

(3) 9C_5

(4) $\frac{9}{5} \cdot {}^8C_4$

Sol. Answer (1, 3, 4)

Number of remaining stations at which train do not halt consecutively = 8

$$S_1 \times S_2 \times S_3 \times S_4 \times S_5 \times S_6 \times S_7 \times S_8$$

$$\text{Required number of ways} = \text{Number of ways of selection of 4 distinct places out of } (8 + 1) \text{ places} = {}^9C_4 = {}^9C_5$$

16. m parallel lines in a plane are intersected by family of n parallel lines. The total number of parallelogram so formed is

(1) ${}^mC_2 \cdot {}^nC_2$

(2) $\frac{m(m-1)(n-1)}{4}$

(3) 150, for $m = 5, n = 6$

(4) 25, for $m = 5, n = 6$

Sol. Answer (1, 3)

A parallelogram is formed by choosing two straight lines from the set of m parallel lines and two straight lines from the set of n parallel lines.

Hence number of parallelogram such formed is

$$= {}^mC_2 \cdot {}^nC_2$$

$$= \frac{m(m-1)}{2} \cdot \frac{n(n-1)}{2}$$

$$= \frac{mn(m-1)(n-1)}{4}$$

for $m = 5, n = 6$

$$\text{Number of parallelogram} = \frac{5 \times 6 \times 4 \times 5}{4} = 150$$

Hence, answer is (1, 3)

17. There are n straight lines in a plane, no two of which are parallel and no three pass through the same point. Their point of intersection are joined. Then the number of fresh lines thus obtained is

(1) ${}^nC_2 \cdot \frac{(n-2)(n-3)}{4}$

(2) $\frac{n(n-1)(n-2)(n-3)}{6}$

(3) $\frac{n(n-1)(n-2)(n-3)}{8}$

(4) $\frac{n(n-1)(n-2)}{8}$

Sol. Answer (1, 3)

Since no two lines are parallel and no three are concurrent, therefore n straight lines intersect at ${}^nC_2 = N$ (let) points. Since two points are required to determine a straight line, therefore, the total number of lines obtained by joining N points NC_2 . But in this each old line has been counted ${}^{n-1}C_2$ times, since on each old line there will be $n-1$ points of intersection made by remaining $(n-1)$ lines. Hence the required number of fresh lines.

$$= {}^NC_2 - n \cdot {}^{n-1}C_2$$

$$= \frac{N(N-1)}{2} - \frac{n(n-1)(n-2)}{2}$$

$$= \frac{{}^nC_2({}^nC_2 - 1)}{2} - \frac{n(n-1)(n-2)}{2}$$

$$= \frac{\frac{n(n-1)}{2} \cdot \left(\frac{n(n-1)}{2} - 2 \right)}{2} - \frac{n(n-1)(n-2)}{2}$$

$$= \frac{n(n-1)(n-2)(n-3)}{8} = {}^nC_2 \cdot \frac{(n-2)(n-3)}{4}$$

Hence answer (1, 3)

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

If n things are arranged in a row, the number of ways in which they can be deranged so that none occupies its original position is

$$n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^n \frac{1}{n!} \right)$$

1. The number of ways of putting 6 letters into 6 addressed envelopes so that exactly 2 letters are in wrong envelopes is

(1) 6C_4 (2) 30 (3) 45 (4) 28

Sol. Answer (1)

$$\text{Required no} = {}^6C_2 = {}^6C_4$$

2. Five boys are to be seated in a row. The number of ways in which 3 boys are not seated in the place specified to them is

(1) 5P_2 (2) 10 (3) 40 (4) 25

Sol. Answer (1)

$${}^5C_2 \times 3! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \right)$$

$$= 10 \times 2 = 20 = {}^5P_2$$

3. The number of ways of putting at least 3 out of 6 letters in wrong envelopes is

(1) 265 (2) 40 (3) 704 (4) 50

Sol. Answer (3)

$${}^6C_3 \times 3! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \right) + {}^6C_4 \times 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right)$$

$$+ {}^6C_5 \times 5! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \right)$$

$$+ {}^6C_6 \times 6! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} \right) = 704$$

Comprehension-II

Five balls are to be placed in three boxes. Each can hold all the five balls. In how many different ways can we place the balls so that no box remains empty, if

1. Balls and boxes are all different

(1) 60 (2) 90 (3) 150 (4) 80

Sol. Answer (3)

As no box is empty, so balls can be of the group 1, 1, 3 or 1, 2, 2.

∴ Required number of arrangement

$$= \frac{5!}{1!1!3!} \times \frac{3!}{2!} + \frac{5!}{1!2!2!} \times \frac{3!}{2!} = \frac{3!}{2!}(20 + 30) = \frac{6 \times 50}{2} = 150$$

Alternate Method

$$\text{Number of ways} = 3^5 - {}^3C_1 2^5 + {}^3C_2 1^5 = 150$$

2. Balls are different but boxes are identical

(1) 15 (2) 25 (3) ${}^5C_1 \times {}^4C_2 \times {}^2C_2$ (4) 30

Sol. Answer (2)

$$\text{Required number of ways} = \frac{5!}{1!1!3!} + \frac{5!}{2!2!1!} = \frac{20}{2} + \frac{30}{2} = 10 + 15 = 25.$$

3. Balls as well as boxes are identical

(1) 5C_3 (2) 2 (3) 1 (4) 6

Sol. Answer (2)

As boxes as well as balls are identical there will be only two arrangements (one in group of 1, 1, 3 and other 1, 2, 2).

Comprehension-III

One and only one straight line can be drawn passing through two given points and we can draw only one triangle through three non-collinear points. By integral coordinates (x, y) of a point we mean both x and y as integers.

1. The number of triangles whose vertices are the vertices of an octagon but none of whose sides happen to be come from the octagon is

(1) 16 (2) 28 (3) 56 (4) 70

Sol. Answer (1)

$$\text{Total number of triangles} = {}^8C_3 = 56.$$

$$\text{Number of triangles whose one side is common to octagon} = 4 \times 8 = 32.$$

$$\text{Number of triangle whose two sides are common to octagon} = 8.$$

$$\text{Required number of triangle} = 56 - 32 - 8 = 16.$$

2. The number of points in the cartesian plane with integral coordinates satisfying the inequalities

$$|x| \leq 4, |y| \leq 4 \text{ and } |x - y| \leq 4 \text{ is}$$

(1) 17

(2) 91

(3) 61

(4) 64

Sol. Answer (3)

$$|x| \leq 4 \Rightarrow -4 \leq x \leq 4$$

$$|y| \leq 4 \Rightarrow -4 \leq y \leq 4$$

$$|x - y| \leq 4 \Rightarrow -4 \leq x - y \leq 4$$

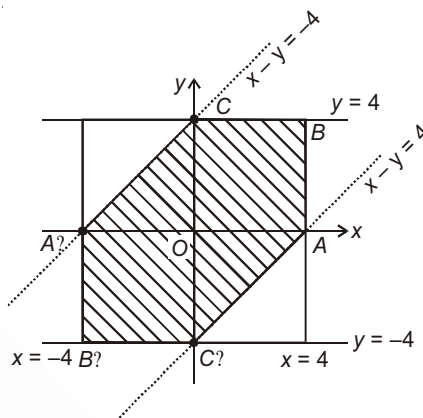
There are $4 \times 4 = 16$ points inside square $OABC$.

Similarly then all $4 \times 4 = 16$ point middle square $OA'B'C'$

there are 6 points each in $\triangle OA'C$ and $\triangle OAC'$.

Also there are 17 points on line $A'A$ and CC' .

Number of solution = Number of integral co-ordinates in shaded region = $16 + 16 + 6 + 6 + 17 = 61$



3. The sides AB , BC , CA of a triangle ABC have a , b and c interior points excluding vertices on them respectively. The number of triangles that can be constructed using these interior points as vertices is

(1) $\frac{ab + bc + ca}{2}$

(2) $\Sigma ab(a + b - 2)$

(3) $\Sigma ab(a + b)$

(4) ${}^{a+b+c}C_3 - ({}^aC_3 + {}^bC_3 + {}^cC_3)$

Sol. Answer (4)

$$\text{Required number of triangle} = {}^{a+b+c}C_3 - {}^aC_3 - {}^bC_3 - {}^cC_3.$$

Comprehension-IV

The number of ways to distribute $m \times n$ different things among n persons equally = $\frac{(mn)!}{(m!)^n}$.

1. The number of ways in which a pack of 52 cards can be distributed among 4 players in a game of bridge is

(1) $\frac{52!}{4!13!}$

(2) $\frac{52!}{4 \times 13!}$

(3) $\frac{52!}{(13!)^4}$

(4) $\frac{50! \times 4!}{13!}$

Sol. Answer (3)

$$\text{Number of ways to distribute 52 cards among 4 players equally} = \frac{52!}{(13!)^4}$$

Hence, option (3) is correct.

2. The number of ways in which a particular player holds all the aces and all of them get equal number is

(1) $\frac{48!}{9!(13!)^3}$

(2) $\frac{48!}{9!(13!)^4}$

(3) $\frac{9 \times 52!}{(13!)^3}$

(4) $\frac{9 \times 52!}{(13!)^4}$

Sol. Answer (1)

When a particular player hold all the aces, then remaining 9 card will other card that player have hence,

$$\text{Required number of ways} = \frac{52!}{9!(13!)^3}$$

Hence, option (1) is correct.

3. The number of ways in which 12, different books can be divided equally in 3 heaps of book is

$$(1) \frac{3 \cdot 12!}{(4!)^3} \quad (2) \frac{12!}{3!(4!)^3} \quad (3) \frac{12!}{3 \cdot (4!)^3} \quad (4) \frac{3! \cdot 12!}{(4!)^3}$$

Sol. Answer (2)

$$12 \text{ books have been divided equally in 3 heaps, then required number of ways} = \frac{12!}{3!(4!)^3}$$

Hence, option (2) is correct.

Comprehension-V

Let a_n denote the number of all n -digit positive integers formed by the digits 0, 1 or both such that no consecutive digits in them are 0. Let b_n = the number of such n -digit integers ending with digit 1 and c_n = the number of such n -digit integers ending with digit 0. [IIT-JEE 2012]

1. Which of the following is correct?

$$(1) a_{17} = a_{16} + a_{15} \quad (2) c_{17} \neq c_{16} + c_{15} \quad (3) b_{17} \neq b_{16} + c_{16} \quad (4) a_{17} = c_{17} + b_{16}$$

Sol. Answer (1)

Let k be the number of 1's and r be the number of zeros and the first digit obviously has to be "1". Clearly

$$a_n = 1 + {}^{n-2}C_1 + {}^{n-3}C_2 + {}^{n-4}C_3 + \dots + {}^kC_r \quad (k \geq r)$$

$$\Rightarrow a_{n-1} = 1 + {}^{n-3}C_1 + {}^{n-4}C_2 + {}^{n-5}C_3 + \dots$$

$$\text{as } a_{n-2} = 1 + {}^{n-4}C_1 + {}^{n-5}C_2 + {}^{n-6}C_3 + \dots$$

$$\text{Clearly, } a_{n-2} + a_{n-1} = a_n$$

$$\text{Hence, } a_{17} = a_{16} + a_{15}$$

2. The value of b_6 is

$$(1) 7 \quad (2) 8 \quad (3) 9 \quad (4) 11$$

Sol. Answer (2)

$$\text{So, } b_n = 1 + {}^4C_1 + {}^3C_2 = 8$$

SECTION - D

Assertion-Reason Type Questions

1. STATEMENT-1 : The number of ways in which six different objects can be divided equally into 2 sets is 10.

and

STATEMENT-2 : The number of ways in which six different objects be distributed equally among two persons is 20.

Sol. Answer (2)

Number of ways of division of six different objects in two sets = $\frac{6!}{3!3!} = 10$

Number of ways of division of six different objects among two persons = $\frac{6!}{3!3!} = 20$

Statement-1 is true, Statement-2 is true but is not a correct explanation.

2. STATEMENT-1 : The exponent of 7 in $^{100}C_{50}$ is 4.

and

STATEMENT-2 : The number of ways in which we can post 5 letters in 12 boxes is 12^5 .

Sol. Answer (4)

Exponent of 7 is 100!

$$= \left[\frac{100}{7} \right] + \left[\frac{100}{7^2} \right] + \left[\frac{100}{7^3} \right]$$

$$= 14 + 2 + 0 = 16$$

Exponent of 7 is 50!

$$= \left[\frac{50}{7} \right] + \left[\frac{50}{7^2} \right] + \left[\frac{50}{7^3} \right]$$

$$= 7 + 1 + 0 = 8$$

$$\text{So, exponent of 7 is } ^{100}C_{50} \text{ is } = 16 - 8 - 8 = 0$$

Hence Statement-1 is false.

3. STATEMENT-1 : The tens digit of $1! + 2! + 3! + 4! + 5! + \dots + 50!$ is 1.

and

STATEMENT-2 : The sum of divisors of $2^4 3^5 5^2 7^3$ is $2^5 \cdot 3^6 \cdot 5^3 \cdot 7^4 - 2 \cdot 3 \cdot 5 \cdot 7$.

Sol. Answer (3)

The sum of divisor of $2^4 3^5 5^2 7^3$ is

$$= (2^0 + 2^1 + 2^2 + \dots + 2^4) (3^0 + 3^1 + \dots + 3^5) (5^0 + 5^1 + 5^2) (7^0 + 7^1 + 7^2 + 7^3)$$

Hence, statement-2 is false.

4. STATEMENT-1 : The range of the function $f(x) = {}^{7-x}C_{x-3}$ is $\{3, 4, 5\}$.

and

STATEMENT-2 : The number of squares which can be formed on a chessboard (8×8) will be 204.

Sol. Answer (4)

$\{3, 4, 5\}$ is the domain of $f(x)$ but not the range

Hence, Statement-1 is false.

5. STATEMENT-1 : The number of ways of choosing 15 couples out of 15 women and 15 men is 1240.

and

STATEMENT-2 : The number of ordered pairs of integers (x, y) which satisfies the equation $y^2 + 6y + x^2 = 4$ is 8.

Sol. Answer (2)

In Statement-1 the required number of ways is

$$\frac{1}{15!} \left({}^{15}C_1 \cdot {}^{15}C_1 \right) \left({}^{14}C_1 \cdot {}^{14}C_1 \right) \left({}^{13}C_1 \cdot {}^{13}C_1 \right) \dots \left({}^1C_1 \cdot {}^1C_1 \right) = \frac{1}{15!} (15!)^2 = 15!$$

In Statement-2

$$y^2 + 6y + 9 = 13 - x^2$$

$$\Rightarrow (y + 3)^2 = 13 - x^2$$

$$\text{When } x = 2 \quad y = 0, -6$$

$$x = -2 \quad y = 0, -6$$

$$x = 3 \quad y = -1, -5$$

$$x = -3 \quad y = -1, -5$$

$\therefore$ Then number of ordered pair of integers is 8.

SECTION - E

Matrix-Match Type Questions

1. How many seven letters words can be formed by using the letters of the word SUCCESS so that?

Column-I

- (A) The two C are together but no two S are together
 (B) Neither two C nor two S are together
 (C) All C are together and S are together and E always comes before U
 (D) U is the starting letter of the word

Column-II

- (p) 60
 (q) 96
 (r) 24
 (s) 12

Sol. Answer A(r), B(q), C(s), D(p)

(A) SUCCESS

_ U _ CC _ E _

If 'S' is placed at the place indicated by _

Then no two 'S' will together

Assuming two 'C' as single unit,

$$\text{Number of arrangement} = 3! \times {}^4C_3 = 6 \times 4 = 24.$$

(B) _ U _ C _ C _ E _

Number of ways when no two 'S' is together

$$= \frac{4!}{2!} \times {}^5C_3 = 4 \times 3 \times 10 = 120$$

Number of arrangement two 'C' are together and no 'S' are together = 24.

∴ Required number = $120 - 24 = 96$.

(C) Assuming all C and S together.

Total number of arrangement = $4! = 24$.

No. of ways when U will con before E = $\frac{24}{2} = 12$.

(D) U

Number of ways when U comes first = $\frac{6!}{3! \cdot 2!} = \frac{6 \times 5 \times 4 \times 3 \times 2}{3 \times 2 \times 2} = 60$

2. Match the following

Column-I

Column-II

- | | |
|--|---------|
| (A) The number of ways in which 35 mangoes can be distributed among 3 boys so that each can have any number of mangoes | (p) 666 |
| (B) The number of triangles whose vertices are at the vertices of a decagon with one side common with the decagon | (q) 60 |
| (C) A box contains 2 white, 3 black and 4 red balls. In how many ways can three balls be drawn from the box if atleast one black ball is to be include in the draw? | (r) 64 |
| (D) A child attempts to open a five disc-lock. He takes 5 seconds to dial a particular number on the disc. If he does so for 5 hours, every day, then the maximum number of days he would take to open the lock. | (s) 28 |

Sol. Answer A(p), B(q), C(r), D(s)

- (A) Required number = ${}^{35+3-1}C_{3-1} = {}^{37}C_2 = 666$.
- (B) Total number of such triangles = $10 \times 6 = 60$.
- (C) Required number of ways = ${}^9C_3 - {}^6C_3 = 84 - 20 = 64$.
- (D) Number of 5 digit number on 5 disc-lock
 $= 10 \times 10 \times 10 \times 10 \times 10 = 10^5$

Number of days = $\frac{10^5}{3600} = 27.77 = 28$ days

3. Match the following

Column-I

Column-II

- | | |
|---|----------|
| (A) The number of squares of any size in a chessboard is | (p) 1092 |
| (B) The number of rectangles of any size which are not squares in a chessboard is | (q) 204 |
| (C) The number of squares of size 3×3 in a chessboard is | (r) 36 |
| (D) The number of rectangles of size 5×4 in a chessboard is | (s) 40 |

Sol. Answer A(q), B(p), C(r), D(s)

(A) Number of squares of any size = $\frac{8(8+1)(17)}{6} = 204$

(B) Number of rectangles of any size which are not squares in a chess board = $\left(\frac{8(8+1)}{2}\right)^2 - 204 = 1092$

(C) Number of squares of size 3×3 in a chessboard
 $= (8 + 1 - 3)(8 + 1 - 3) = 6^2 = 36$.

(D) Number of rectangles of size 5×4 in a chess board
 $= 2 \times (5 \times 4) = 40$

4. Match the following

Column I

Column II

(A) There are 8 Hindi novels and 6 English novels. 4 Hindi novels and

(p) $\frac{5 \cdot (8!)}{56}$

3 English novels are selected and arranged in a row such that they are alternate then no. of ways is

(B) Number of arrangements of letters of the word CONCRETE such that no two vowels are together is

(q) $5(8!)$

(C) There are 10 AC in a hall. In how many ways they can be operated such that atleast 4 AC is on?

(r) 848

(D) $A = \{1, 2, 3, \dots, 10, 11\}$, Number of subsets of A having atleast 6 elements is equal to

(s) 1024

(t) ${}^8P_4 \cdot {}^6P_3$

Sol. Answer A(q, t), B(p), C(r), D(s)

(A) Books are in alternate order, then they must be in the order

$$H_1 E_1 H_2 E_2 H_3 E_3 H_4$$

Total no. of ways = ${}^8C_4 \times {}^6C_3 \times 4! \times 3! = 5 \cdot (8!)$

(B) CONCRETE

vowels are O, E, E

_ C _ N _ C _ R _ T _

If vowels are placed at the place indicated by _ then no two vowels will be together

Now required no. of ways = $\frac{5!}{2!} \times {}^6C_3 \times \frac{3!}{2!}$

$$\frac{5!}{2!} \times \frac{6 \times 5 \times 4}{3!} \times \frac{3!}{2!} = 5(6!) = \frac{5(8!)}{56}$$

(C) Required no. of ways = ${}^{10}C_4 \times 2^6 = 8! / 3$

(D) Required no. of subsets with the given condition = ${}^{11}C_6 + {}^{11}C_7 + {}^{11}C_8 + {}^{11}C_9 + {}^{11}C_{10} + {}^{11}C_{11}$

$$= \frac{1}{2} [{}^{11}C_0 + {}^{11}C_1 + \dots + {}^{11}C_{11}] = 2^{10} = 1024$$

5. Match the entries of Column-I with those of Column-II.

Column I**Column II**

- | | |
|--|--------|
| (A) The value of x satisfying the inequality ${}^{10}C_{x-1} > 2 \cdot {}^{10}C_x$ is | (p) 9 |
| (B) A person wishes to make up as many different parties as he can out of his 20 friends such that each party consists of the same number of persons. The number of friends he should invite at a time is less than or equal to | (q) 7 |
| (C) There are four letters and four directed envelopes. The number of ways in which all the letters can be put in the wrong envelopes is | (r) 8 |
| (D) If a denotes the number of permutations of $x + 2$ things taken all at a time, b the number of permutations of x things taken 11 at a time and c the number of permutations of $x - 11$ things taken all at a time such that $a = 182bc$, then the value of x is greater than or equal to | (s) 10 |
| | (t) 12 |

Sol. Answer A(p, r, s), B(s, t), C(p), D(p, q, r, s, t)

(A) ${}^{10}C_{x-1} > 2 \cdot {}^{10}C_x$

$$\Rightarrow \frac{10!}{(x-1)!(10-x+1)!} > 2 \cdot \frac{10!}{(10-x)!x!}$$

$$\Rightarrow \frac{1}{11-x} > \frac{2}{x} \quad x \in I^+$$

$$\Rightarrow x > 22 - 2x$$

$$3x > 22 \Rightarrow x > \frac{22}{3}$$

$$\therefore x = 8, 9, 10$$

- (B) Suppose he invites r friends at a time, then the total number of parties is ${}^{20}C_r$. We have to find maximum value of ${}^{20}C_r$, which is for $r = 10$

$\therefore$ He should invite 10 friends at a time in order to form the maximum number of parties.

- (C) Required number of ways

$$= 4! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right]$$

$$= 12 - 4 + 1 = 9$$

- (D) We have, $a = {}^{x+2}P_{x+2} = (x+2)!$

$$b = {}^xP_{11} = \frac{x!}{(x-11)!}$$

$$c = {}^{x-11}P_{x-11} = (x-11)!$$

$$\text{Now, } a = 182bc$$

$$\Rightarrow (x+2)! = 182 \cdot \frac{x!}{(x-11)!} \cdot (x-11)!$$

$$\Rightarrow (x+2)! = 182 \times x!$$

$$\Rightarrow \boxed{x=12}$$

6. Match the entries of Column-I with those of Column-II

Column I**Column II**

- | | |
|---|------------------|
| (A) The number of non-negative integral solutions of $2x + y = 11$ is less than or equal to | (p) 6 |
| (B) The number of ways in which 7 identical rings can be put on 5 fingers of a hand is greater than or equal to if each finger has atleast one ring | (q) 9 |
| (C) If the number of selections of r things as well as $10 + r$ things out of 30 different things be equal then r is less than or equal to | (r) 10 |
| (D) The number of ways in which 6 identical things can be distributed among 3 persons so that each gets at least one, is | (s) 16
(t) 15 |

Sol. Answer A(p, q, r, s, t), B(p, q, r, t), C(r, s, t), D(r)

(A) Clearly $x = 0, 1, 2, 3, 4, 5$ Let $x = k$ then $0 \leq k \leq 5$

When $x = k, y = 11 - 2k$

The number of non-negative integral solutions

= The number of ways to distribute $(11 - 2k)$ identical things to one person.

$$= {}^{11-2k+1-1}C_{1-1} = 1$$

$$\sum_{k=0}^5 1 = 6$$

(A) 7 identical rings can be put on 5 fingers. Such that each finger has at least one ring. The required number of ways = ${}^{7-1}C_{5-1} = {}^6C_4 = 15$

(B) We have ${}^{30}C_r = {}^{30}C_{10+r}$

$$\Rightarrow r + 10 + r = 30$$

$$r = 10$$

$$\Rightarrow \boxed{r=10}$$

(D) Number of ways to distribute 6 identical things among 3 persons.

$$= {}^{6-1}C_{3-1} = {}^5C_2$$

$$= \frac{5 \times 4}{2 \times 1} = 10$$

7. Consider all possible permutations of the letters of the word ENDEANOEL. Match the Statements/Expressions in **Column I** with the Statements/Expressions in **Column II**. [IIT-JEE 2008]

Column I**Column II**

- | | |
|--|--------------------|
| (A) The number of permutations containing the word ENDEA is | (p) 5! |
| (B) The number of permutations in which the letter E occurs in the first and the last positions is | (q) $2 \times 5!$ |
| (C) The number of permutations in which none of the letters D, L, N occurs in the last five positions is | (r) $7 \times 5!$ |
| (D) The number of permutations in which the letters A, E, O occur only in odd positions is | (s) $21 \times 5!$ |

Sol. Answer A(p), B(s), C(q), D(q)

(A) If ENDEA is fixed word then assume this as a single letter

Total letters = 5, total number of arrangement = 5!

(B) If E is at first and last places then

$$\text{Total permutation} = \frac{7!}{2!} = 21 \times 5!$$

(C) If D, L, N is not in last five position



$$\text{Total permutation} = \frac{4!}{2!} \times \frac{5!}{3!} = 2 \times 5!$$

(D) Total odd position = 5

$$\text{Permutation of } AEEEE \text{ are } \frac{5!}{3!}$$

Total even position = 4

$$\text{Permutation of } N, N, D, L = \frac{4!}{2!}$$

$$\text{Hence total permutation} = \frac{5!}{3!} \times \frac{4!}{2!} = 2 \times 5!$$

8. In a high school, a committee has to be formed from a group of 6 boys $M_1, M_2, M_3, M_4, M_5, M_6$ and 5 girls G_1, G_2, G_3, G_4, G_5 .

- Let α_1 be the total number of ways in which the committee can be formed such that the committee has 5 members, having exactly 3 boys and 2 girls.
- Let α_2 be the total number of ways in which the committee can be formed such that the committee has at least 2 members, and having an equal number of boys and girls.
- Let α_3 be the total number of ways in which the committee can be formed such that the committee has 5 members, at least 2 of them being girls.
- Let α_4 be the total number of ways in which the committee can be formed such that the committee has 4 members, having at least 2 girls and such that both M_1 and G_1 are NOT in the committee together.

[JEE(Advanced)-2018]

Column-I

- The value of α_1 is
- The value of α_2 is
- The value of α_3 is
- The value of α_4 is

Column-II

- 136
- 189
- 192
- 200
- 381
- 461

The correct option is :

- (1) A(s); B(u); C(q); D(p)
- (2) A(p); B(s); C(q); D(r)
- (3) A(s); B(u); C(t); D(q)
- (4) A(s); B(q); C(r); D(p)

Sol. Answer (3)

$$(i) \alpha_1 = {}^6C_3 \times {}^5C_2 = 20 \times 10 = 200$$

$$\begin{aligned}
 \text{(ii)} \quad \alpha_2 &= {}^6C_1 \times {}^5C_1 + {}^6C_2 \times {}^5C_2 + {}^6C_3 \times {}^5C_3 + {}^6C_4 \times {}^5C_4 + {}^6C_5 \times {}^5C_5 \\
 &= 30 + 150 + 200 + 75 + 6 = 461 \\
 \text{(iii)} \quad \alpha_3 &= {}^5C_2 \times {}^6C_3 + {}^5C_3 \times {}^6C_2 + {}^5C_4 \times {}^6C_1 + {}^5C_5 \\
 &= 200 + 150 + 30 + 1 = 381 \\
 \text{(iv)} \quad \alpha_4 &= ({}^5C_2 \times {}^6C_3 - {}^4C_1 \times {}^5C_1) + ({}^5C_3 \times {}^6C_1 - {}^4C_2 \times {}^5C_0) \\
 &= (150 - 20) + (60 - 6) + 5 \\
 &= 130 + 60 - 1 = 190 - 1 = 189
 \end{aligned}$$

SECTION - F

Integer Answer Type Questions

1. An eight digit number is formed from 1, 2, 3, 4 such that product of all digits is always 3072, the total number of ways is $(23 \cdot {}^8C_k)$, where the value of k is _____.

Sol. Answer (3) or (5)

$$3072 = 2^{10} \times 3$$

2	2	2	2	4	4	4	3
---	---	---	---	---	---	---	---

$$\text{No. of ways} = \frac{8!}{4!3!}$$

2	2	2	2	4	4	4	3
---	---	---	---	---	---	---	---

$$\text{No. of ways} = \frac{8!}{4!2!}$$

1	1	4	4	4	4	4	3
---	---	---	---	---	---	---	---

$$\text{No. of ways} = \frac{8!}{5!2!}$$

$$\text{Total no. of ways} = \frac{8!}{4!3!} + \frac{8!}{4!2!} + \frac{8!}{5!2!} = \frac{23 \cdot 8!}{5!3!} = 23 \cdot {}^8C_3 = 23 \cdot {}^8C_5$$

2. The number of ways of distributing six identical mathematics books and six identical physics books among three students such that each student gets atleast one mathematics book and atleast one physics book is $\frac{5 \cdot 5!}{k}$, then k is _____.

Sol. Answer (6)

$$\begin{aligned}
 \text{Required no. of ways} &= {}^{6-1}C_{3-1} \cdot {}^{6-1}C_{3-1} = 10 \times 10 = 100 = \frac{5! \times 5}{6} \\
 \therefore k &= 6
 \end{aligned}$$

3. There are red, green and white identical balls, each being 10 in number. The number of selections of 10 balls in which the number of red balls is double the number of green balls is _____.

Sol. Answer (4)

Since all the balls are identical

Let x green balls be selected, y white ball is selected

So, red balls will be $2x$.

$$\text{Now, } x + 2x + y = 10$$

$$\Rightarrow 3x + y = 10$$

Clearly $x = 0, 1, 2, 3$ i.e., 4 possible selections.

Corresponding values of y are 10, 7, 4, 1

4. If the number of ways in which a lawn-tennis mixed double be made from seven married couples if no husband and wife play in the same set is k , then greatest prime divisor of k is _____.

Sol. Answer (7)

$$\text{Required no. of ways } k = {}^7C_2 \cdot {}^5C_2 \cdot 2 = 420$$

Greatest prime divisor of $k = 7$

5. Let $n_1 < n_2 < n_3 < n_4 < n_5$ be positive integers such that $n_1 + n_2 + n_3 + n_4 + n_5 = 20$. Then the number of such distinct arrangements $(n_1, n_2, n_3, n_4, n_5)$ is _____. **[JEE(Advanced)-2014]**

Sol. Answer (7)

$$n_1 + n_2 + n_3 + n_4 + n_5 = 20$$

$$\text{Maximum of } n_5 = 10. \quad n_1 = 1, n_2 = 2, n_3 = 3, n_4 = 4$$

$$n_5 = 09. \quad n_1 = 1, n_2 = 2, n_3 = 3, n_4 = 5$$

$$n_5 = 08. \quad n_1 = 1, n_2 = 2, n_3 = 3, n_4 = 6$$

$$\text{or } n_3 = 4, n_4 = 5$$

$$n_5 = 07. \quad n_1 = 1, n_2 = 2, n_3 = 4, n_4 = 6$$

$$\text{or } n_2 = 3, n_3 = 4, n_4 = 5$$

$$n_5 = 06. \quad n_1 = 2, n_2 = 3, n_3 = 4, n_4 = 5$$

Total number of ways = 7.

6. Let $n \geq 2$ be an integer. Take n distinct points on a circle and join each pair of points by a line segment. Colour the line segment joining every pair of adjacent points by blue and the rest by red. If the number of red and blue line segments are equal, then the value of n is _____. **[JEE(Advanced)-2014]**

Sol. Answer (5)

Number of line joining adjacent points = n

$$n = {}^nC_2 - n$$

$$2n = {}^nC_2$$

$$2n = \frac{n(n-1)}{2}$$

$$n = 0 \text{ or } n = 5$$

7. Let n be the number of ways in which 5 boys and 5 girls can stand in a queue in such a way that all the girls stand consecutively in the queue. Let m be the number of ways in which 5 boys and 5 girls can stand

in a queue in such a way that exactly four girls stand consecutively in the queue. Then the value of $\frac{m}{n}$ is

[JEE(Advanced)-2015]

Sol. Answer (5)

$$n = 6 \cdot 5$$

$$m = {}^5C_4 \times {}^6P_2 \times 4! \times 5!$$

$$\frac{m}{n} = 5$$

8. Words of length 10 are formed using the letters $A, B, C, D, E, F, G, H, I, J$. Let x be the number of such words where no letter is repeated; and let y be the number of such words where exactly one letter is repeated twice and no other letter is repeated. Then, $\frac{y}{9x} =$ [JEE(Advanced)-2017]

Sol. Answer (5)

Clearly $x = 10!$

For calculating y , one letter has to be excluded, this can be done in 10 ways.

Out of the remaining 9 letters, one of them is to be repeated

$$y = {}^{10}C_1 \times {}^9C_1 \times \frac{10!}{2!} = 5 \times 9 \times 10!$$

$$\therefore \frac{y}{9x} = \frac{5 \times 9 \times 10!}{9 \times 10!} = 5$$

9. The number of 5 digit numbers which are divisible by 4, with digits from the set $\{1, 2, 3, 4, 5\}$ and the repetition of digits is allowed, is _____. [JEE(Advanced)-2018]

Sol. Answer (625)

Last two digits can be 12, 24, 32, 44 and 52

Hence, number of 5 digit numbers $= 5 \times 5^3 = 625$

10. Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one-one functions from X to Y and β is the number of onto functions from Y to X , then the value of $\frac{1}{5!}(\beta - \alpha)$ is _____. [JEE(Advanced)-2018]

Sol. Answer (119.00)

α = Number of one-one functions from X to Y

β = Number of onto functions from Y to X

= Make 5 groups out of 7 elements of Y and permute these groups to 5 elements of X

$$= \frac{{}^7P_5}{{}^3P_1} \times {}^5P_5 + \frac{{}^7P_4}{{}^2P_2} \times {}^4P_4 + \frac{{}^7P_3}{{}^1P_1} \times {}^3P_3 + \frac{{}^7P_2}{{}^1P_1} \times {}^2P_2 + \frac{{}^7P_1}{{}^1P_1} \times {}^1P_1$$

$$\Rightarrow \frac{1}{5!}(\beta - \alpha) = \frac{{}^7P_5}{{}^3P_1} \times \frac{{}^5P_5}{5!} + \frac{{}^7P_4}{{}^2P_2} \times \frac{{}^4P_4}{4!} + \frac{{}^7P_3}{{}^1P_1} \times \frac{{}^3P_3}{3!} + \frac{{}^7P_2}{{}^1P_1} \times \frac{{}^2P_2}{2!} + \frac{{}^7P_1}{{}^1P_1} \times \frac{{}^1P_1}{1!} - \frac{{}^7P_5}{{}^3P_1} = 7[5 + 15 - 3] = 119$$

SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1 : Number of ways of distributing 20 different things equally among 5 persons $= 20!/4!^5$.
STATEMENT-2 : Number of ways of distributing 20 different things equally among 5 groups $= 20!/4!^5 \cdot 5!$.
STATEMENT-3 : Number of ways of distributing 20 different things in 5 groups such that in three group there are 6 things each and in remaining two groups there is 1 thing each $= \frac{20!}{(6!)^3}$.

(1) T F T

(2) T T F

(3) F T T

(4) T T T

Sol. Answer (2)

Statement-1: True

Division of groups when order is important = $\frac{20!}{(4!)^5}$

Statement-2: True

When order is not important = $\frac{20!}{(4!)^5 \times 5}$

Statement-3: False

No. of ways of division into groups = $\frac{20!}{(6!)^3 \times 5}$

2. STATEMENT-1 : There are 12 points in a plane of which only 5 are collinear, then the number of straight lines obtained by joining these points in pairs is ${}^{12}C_2 - {}^5C_2$.

STATEMENT-2 : ${}^{n+1}C_r - {}^{n-1}C_{r-1} = {}^nC_r + {}^nC_{r-2}$.

STATEMENT-3 : $2n$ persons may be seated at two round tables, n person being seated at each, in $\frac{(2n)!}{n^2}$ different ways.

(1) F T F

(2) T T F

(3) F F T

(4) F T T

Sol. Answer (3)Statement-1 : Number of lines = ${}^{12}C_2 - {}^5C_2 + 1$

Hence false.

Statement-2 : ${}^{n+1}C_r - {}^{n-1}C_{r-1} = ({}^nC_r + {}^nC_{r-1}) - {}^{n-1}C_{r-1}$
 $= [{}^nC_r + ({}^{n-1}C_{r-1} + {}^{n-1}C_{r-2})] - {}^{n-1}C_{r-1}$
 $= {}^nC_r + {}^{n-1}C_{r-2}$

Hence false.

Statement-3: Required number of ways

$$= {}^{2n}C_n \cdot (n-1)! (n-1)! = \frac{2n!}{n!n!} \cdot (n-1)! (n-1)! \\ = \frac{(2n)!}{n^2}$$

Hence true.

3. STATEMENT-1 : The product of r consecutive integers is always divisible by $r!$.

STATEMENT-2 : The number of words formed by utilizing all letters of the word ASSIST in which S's are alternate, is 12.

STATEMENT-3 : The number of words which can be formed out of the letters a, b, c, d, e, f taken 3 together, each word containing one vowel at least is 16.

(1) T F T

(2) T T F

(3) F T T

(4) T T T

Sol. Answer (2)

$$\text{Statement-1: } n(n-1)(n-2) \dots (n-r+1) = \frac{n(n-1) \dots (n-r+1)(n-r)(n-r-1) \dots 3 \cdot 2 \cdot 1}{(n-r)!}$$

$$= r! {}^nC_r$$

Statement-2: The required number of ways is $3! \times 2 = 12$

$$\text{Statement-3: The number of words is } = ({}^2C_1 \times {}^4C_2 + {}^2C_2 \times {}^4C_1)3!$$

$$= (2 \times 6 + 1 \times 4)6 = 96$$

4. STATEMENT-1 : The total number of combinations of n things by taking some or all is 2^n .

STATEMENT-2 : 5 identical balls can be distributed among 10 identical boxes in only one way if not more than one ball can go into a box.

STATEMENT-3 : In the permutations of n things taken r at a time, the number of permutations in which m particular things occur together is ${}^{n-m}P_{r-m} \cdot {}^rP_m$.

(1) F T F

(2) F F F

(3) T F T

(4) F T T

Sol. Answer (4)

Statement-1: The total number of combinations of n things by taking some or all

$$= {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n - 1$$

Hence false.

Statement-2: Obviously there is only one way of doing it.

Statement-3: Total number of permutation is

$$= {}^{n-m}C_{r-m} \cdot r! = \frac{(n-m)!}{(n-r)!(r-m)!} \cdot r!$$

$$= {}^{n-m}P_{r-m} \cdot {}^rP_m$$

Hence, true

SECTION - H

Aakash Challengers Questions

1. The number of ways can 14 identical toys distributed among three boys so that each one gets atleast one toy and no two boys get equal number of toys is _____.

Sol. Number of ways to distribute at least one toy to each $= {}^{14-1}C_{3-1} = {}^{13}C_2 = 78$

If toys are distributed in the following way then two will get equal number of toys

No. of ways

$$1 \ 1 \ 12 \quad \frac{3!}{2!} = 3 \text{ ways}$$

$$2 \ 2 \ 10 \quad 3 \text{ ways}$$

$$3 \ 3 \ 8 \quad 3 \text{ ways}$$

$$4 \ 4 \ 6 \quad 3 \text{ ways}$$

$$5 \ 5 \ 4 \quad 3 \text{ ways}$$

$$6 \ 6 \ 2 \quad 3 \text{ ways}$$

$\therefore$ Required number of ways $= 78 - 18 = 60$.

2. The number of ordered triplets of positive integers which satisfy the inequality $20 \leq x + y + z \leq 50$ is p , then

$$\frac{p-631}{100} \text{ is equal to } \underline{\hspace{2cm}}.$$

Sol. $x + y + z = 19$

$$20 \leq x + y + z \leq 50$$

$$\begin{aligned} P &= {}^{20-1}C_{3-1} + {}^{21-1}C_{3-1} + \dots + {}^{50-1}C_{3-1} \\ &= {}^{19}C_2 + {}^{20}C_2 + \dots + {}^{49}C_2 \\ &= {}^{19}C_3 + {}^{20}C_2 + \dots + {}^{49}C_2 - {}^{19}C_3 \\ &= {}^{50}C_3 - {}^{19}C_3 = 19600 - 969 = 18631. \end{aligned}$$

$$\text{Now, } \frac{p-631}{100} = \frac{18000}{100} = 180$$

3. The number of 6 digit numbers that contains 6 exactly once is $\underline{\hspace{2cm}}$.

Sol. If repetition of digits is not allowed.

When 6 is at lakh's place, remaining five places can be filled by remaining digits 0, 1, 2, 3, 4, 5, 7, 8, 9 in 9P_5 ways. When 6 is at any other place, the lakh's place can be filled by 8 different ways and remaining five places in 9P_5 ways.

Hence total number of six digit numbers that contains 6 exactly one

$$\begin{aligned} &= {}^9P_5 + 8 \cdot {}^9P_5 = 9 \cdot {}^9P_5 \\ &= 9 \times 9 \times 8 \times 7 \times 6 = 27,216. \end{aligned}$$

If repetition of digits is permissible, then total number of 6 digit numbers with the given condition is $9^5 + 5 \times 8 \times 9^4$.

4. The number of five digits can be made with the digits 1, 2, 3 each of which can be used atmost thrice in a number is $\underline{\hspace{2cm}}$.

Sol. We have the digits 1, 1, 1, 2, 2, 2, 3, 3, 3 to make numbers of five digits. The digits will be as follows

(i) **Case I :** Three identical, one pair

$$\text{The number of selections of three identical digits, one pair} = {}^3C_1 \times {}^2C_1$$

$$\text{Corresponding to each selection, the number of numbers that can be made} = \frac{5!}{3!2!}$$

$\therefore$ The total number of numbers of three identical digits and one pair

$$= {}^3C_1 \cdot {}^2C_1 \cdot \frac{5!}{3!2!} = 3 \times 2 \times \frac{5 \times 4}{2} = 60$$

(ii) **Case II :** Three identical, two different

$$\text{The number of selections of three identical digits and two different digits} = {}^3C_1 \times {}^2C_2$$

$\therefore$ The total number of numbers of three identical digits and two different digits

$$= {}^3C_1 \cdot {}^2C_2 \cdot \frac{5!}{3!} = 60$$

(iii) **Case III** : The number of selections of two pairs and one different digit = ${}^3C_2 \times {}^1C_1$

$\therefore$ The total number of numbers of two pairs and one different digit

$$= {}^3C_2 \cdot {}^1C_1 \cdot \frac{5!}{2!2!} = 3 \times 30 = 90$$

$\therefore$ The required number of numbers = $60 + 60 + 90 = 210$

5. Find the number of proper divisors of the number 8820. Also find the sum of the odd proper divisors of the number.

Sol. Here $8820 = 882 \times 10 = 18 \times 49 \times 10$

$$= 2 \times 3^2 \times 7^2 \times 2 \times 5$$

$$= 2^2 \times 3^2 \times 5^1 \times 7^2$$

$\therefore$ The required number of factors

= The total number of selections from

$(2, 2), (3, 3), (5), (7, 7)$

$$= (2+1)(2+1)(1+1)(2+1) - 2$$

$$= 3 \times 3 \times 2 \times 3 = 54 - 2 = 52$$

The required sum = $(3^0 + 3^1 + 3^2)(5^0 + 5^1)(7^0 + 7^1 + 7^2) - 1$

$$= (1 + 3 + 9)(1 + 5)(1 + 7 + 49) - 1$$

$$= 13 \times 6 \times 57 = 4446 - 1 \text{ (} 2^n \text{ as a factor and 1 as a odd divisor are to be excluded)}$$

$$= 4445$$

6. Let n and k be positive integers such that $n \geq \frac{k(k+1)}{2}$. Find the number of solutions $(x_1, x_2, x_3, \dots, x_k)$,

$x_1 \geq 1, x_2 \geq 2, \dots, x_k \geq k$, all integers satisfying the condition $x_1 + x_2 + x_3 + \dots + x_k = n$.

Sol. Let $x = y_1 + 1, x_2 = y_2 + 2, \dots, x_k = y_k + k$

Putting these in the equation,

$$(y_1 + 1) + (y_2 + 2) + (y_3 + 3) + \dots + (y_k + k) = n$$

$$\Rightarrow y_1 + y_2 + y_3 + \dots + y_k = n - \frac{k(k+1)}{2} = m \text{ (Let) where } m \geq 0 \quad \dots(i)$$

Now, number of non-negative integral solution of (i) is, $= {}^{m+k-1}C_{k-1}$

The above relation is also the number of solution of $x_1 + x_2 + x_3 + \dots + x_k = n$

Hence, required solution is ${}^{m+k-1}C_{k-1}$ where $m = n - \frac{k(k+1)}{2} = \frac{(m+k-1)}{(k-1)!m!}$

7. $2n$ objects of each of three kinds are given to two persons, so that each person gets $3n$ objects. Prove that this can be done in $3n^2 + 2n + 1$ ways.

Sol. If the first person gets $n - p$ ($P \in N_0$) objects of first kind then the person can get q to $2n$ objects of the second kind. The remaining ones are objects of the third kind. The sum is

$$\sum_{p=0}^n (2n - p + 1) = (2n+1)(n+1) - \frac{n(n+1)}{2}$$

If the person gets $n + q$ objects of the first kind, then the person gets 0 to $(2n - q)$ objects of the second kind, since the person gets $3n$ objects altogether. The sum is

$$\sum_{q=1}^N (2n - q + 1) = n(2n + 1) - \frac{n(n + 1)}{2}$$

Hence the required number of ways is

$$\begin{aligned} &= (2n + 1)(n + 1) + n(2n + 1) - n(n + 1) \\ &= 2n^2 + 2n + n + 1 + 2n^2 + n - n^2 - n \\ &= 3n^2 + 3n + 1 \end{aligned}$$

8. How many 5-digit numbers from the digits $\{0, 1, \dots, 9\}$ have?

- (i) Strictly increasing digits (ii) Strictly increasing or decreasing digits
(iii) Increasing digits (iv) Increasing or decreasing digits

Sol. (i) Number of strictly increasing digits $= {}^9C_5 = 126$. (Any five numbers chosen from 1 to 9 can be arranged in one way only).

(ii) Number of strictly increasing or decreasing digits $= 126 + ({}^{10}C_5) = 378$

(iii) Number of increasing digits $= {}^9C_1(1) + {}^9C_2(4) + {}^9C_3(6) + {}^9C_4(4) + {}^9C_5(1) = 1287$

(iv) Number of increasing or decreasing digits $= 1287 + 1287 + {}^9C_1(1) + {}^9C_2(3) + {}^9C_3(3) + {}^9C_4(1)$
 $= 3288$

9. The number of triangles that can be formed from a regular polygon of $2n + 1$ sides such that the centre of the polygon lies inside the triangle is

- (1) $\frac{n}{2}(n + 1)$ (2) $\frac{n^2 + 3n}{2}$ (3) $\frac{n(n + 1)(2n + 1)}{6}$ (4) $\left(\frac{n(n + 1)}{2}\right)^2$

Sol. Answer (3)

Mark all the vertices $1, 2, 3, \dots, (2n + 1)$. If k and m are the two vertices of the triangle, draw two diameters from k and m , then we have four arcs. Then to satisfy the required condition third vertex should lie on the arc diametrically opposite to the arc ' km '. So now we have to find the points lying on that arc. We can notice that there will be $a = |k - m|$ points on that arc.

If a comes out to be greater than n , then a should be treated as $(2n + 1 - |k - m|)$

Choose the first vertex in $(2n + 1)$ ways. Suppose $(2n + 1)^{\text{th}}$ is the chosen vertex, then if we choose 1 as next vertex then for the third vertex we have only one option, as

$$k - m = 2n \Rightarrow a = 1$$

if second vertex is 2, then 2

if n , then n

if $n + 1$, then n , if $2n$ then 1

$$\text{total} = \frac{2n(n + 1)}{2} = n(n + 1)$$

So, total number of triangle $= n(n + 1)(2n + 1)$

But since order is not important so total no. of triangle $= \frac{n(n + 1)(2n + 1)}{3!}$

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13. Determine the number of ordered pairs of positive integers (a, b) such that the least common multiple of a and b is $2^3 \cdot 5^7 \cdot 11^{13}$.

Sol. Both a and b are factors of $2^3 \cdot 5^7 \cdot 11^{13}$ and so $a = 2^x \cdot 5^y \cdot 11^z$ and $b = 2^s \cdot 5^t \cdot 11^u$ for some non-negative integers x, y, z, s, t, u because $2^3 \cdot 5^7 \cdot 11^{13}$ is the least common multiple. $\max\{x, s\} = 3$, $\max\{y, t\} = 7$, and $\max\{z, u\} = 13$.

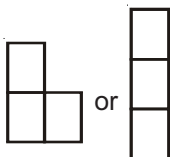
Hence (x, s) can be $(0, 3), (1, 3), (2, 3), (3, 3), (3, 2), (3, 1)$ or $(3, 0)$. So there are 7 choices for $\{x, s\}$

Similarly $\{y, t\}$ can be in number 15 and $\{z, u\}$ can be in number 27.

Required number of ordered pairs of positive integers $= 7 \times 15 \times 27 = 2835$

14. In how many ways we can choose 3 squares on a chess board such that one of the squares has its two sides common to other two squares?

Sol. Either we have to choose



- (i) Every square of (2×2) will contribute four shaped  figures by removing any one square out of it.

Number of ways to choose these squares $= 7 \times 7 = 49$

So, total of  shaped figures $= 49 \times 4 = 196$

- (ii) In every line it is possible to have 6  shaped figures.

So, total number of such figures $= 6 \times 8 \times 2 = 96$

$\therefore$ Total number of cases $= 196 + 96 = 292$

15. Let $A = \{1, 2, 3, \dots, n\}$, if a_i is the minimum element of the set A_i (where A_i denotes the subset of A containing exactly three elements) and X denotes the set of A_i 's, then evaluate $\sum_{A_i \in X} a_i$.

Sol. Since we can take three elements out of n in $\frac{n}{3}$ ways, x will contain nC_3 subsets. There are exactly ${}^{n-1}C_2$ elements of x having 1 as least element, exactly ${}^{n-2}C_2$ elements of x having 2 as least element and so on In general, there are exactly ${}^{n-r}C_2$ elements of x having r ($1 \leq r \leq n-2$) as least elements.

Thus,

$$\sum_{A_i \in X} \min(a_i) = 1 \cdot {}^{n-1}C_2 + 2 \cdot {}^{n-2}C_2 + 3 \cdot {}^{n-3}C_2 + \dots + (n-2) \cdot {}^2C_2$$

$$= \frac{1}{24}(n+1)n(n-1)(n-2)$$

$$= {}^{n+1}C_4$$

